

Global Considerations in Kidney Disease: Africa

Mogamat Razeen Davids | Mogamat-Yazied Chothia | Robert Kalyesubula

CHAPTER OUTLINE

THE RISING BURDEN OF NONCOMMUNICABLE DISEASES IN AFRICA, 2225	ACUTE KIDNEY INJURY, 2228
HUMAN RESOURCES FOR NEPHROLOGY, 2228	CHRONIC KIDNEY DISEASE, 2232
	KIDNEY FAILURE AND KIDNEY REPLACEMENT THERAPY, 2239
	CONCLUSION, 2241

KEY POINTS

- The growing burden of kidney disease in Africa is driven by high rates of infectious disease, noncommunicable disease, pregnancy-related disease, and injuries.
- In most African countries, financial and human resources are inadequate to meet this challenge, and many governments have not yet recognized chronic kidney disease as a national health priority.
- Most patients with kidney failure cannot access kidney replacement therapy, and the lack of access to acute dialysis results in many preventable deaths due to acute kidney injury.
- Examples of initiatives that provide hope for the future include the training of nephrology health care professionals via the International Society of Nephrology and International Society for Peritoneal Dialysis Fellowship programs, increased awareness of kidney disease resulting from World Kidney Day activities, and the Saving Young Lives project, which is improving the outcomes for African patients with acute kidney injury.

Africa is the world's second largest and second most populous continent including 54 fully recognized states and diverse ethnic groups, religions, and languages. Its population in mid-2022 was estimated at 1.42 billion and is projected to reach 2.48 billion by 2050 (Fig. 75.1).^{1,2} Half of the world's population growth is expected to be concentrated in just eight countries including five countries from Africa: Nigeria, the Democratic Republic of the Congo, Egypt, Ethiopia, and Tanzania.²

The African population is young, with 40% being younger than the age of 15 years and only 3% being 65 years or older (Fig. 75.2), compared with the world averages of 25% and 10%, respectively.^{1,3} The average lifetime births per woman is 4.3.^{1,3} Child marriages and adolescent childbearing are common and have serious health consequences, as well as limiting educational and employment opportunities.

Economic growth in sub-Saharan Africa slowed to 3.6% in 2022, from 4.1% in 2021.⁴ Consumer price inflation hit a 14-year high of 9.2% in 2022, fueled by rising food and energy prices, as well as weaker currencies. Health spending in 2020

averaged 5.4% of gross domestic product, being highest in Lesotho (11.8%) and lowest in Djibouti (2.0%) (Fig. 75.3).⁵

Africa labors under the quadruple burdens of communicable, noncommunicable, perinatal and maternal, and injury-related disorders. These factors are driving the current epidemic of kidney disease. Acute kidney injury (AKI) often develops in the setting of infectious diseases, complicated pregnancies, or injuries. Chronic kidney disease (CKD) may be related to HIV or other infections and is a frequent complication of noncommunicable diseases (NCDs) like diabetes mellitus and hypertension.

THE RISING BURDEN OF NONCOMMUNICABLE DISEASES IN AFRICA

As a result of rising incomes, population growth and aging, globalization, rapid urbanization, and Westernized

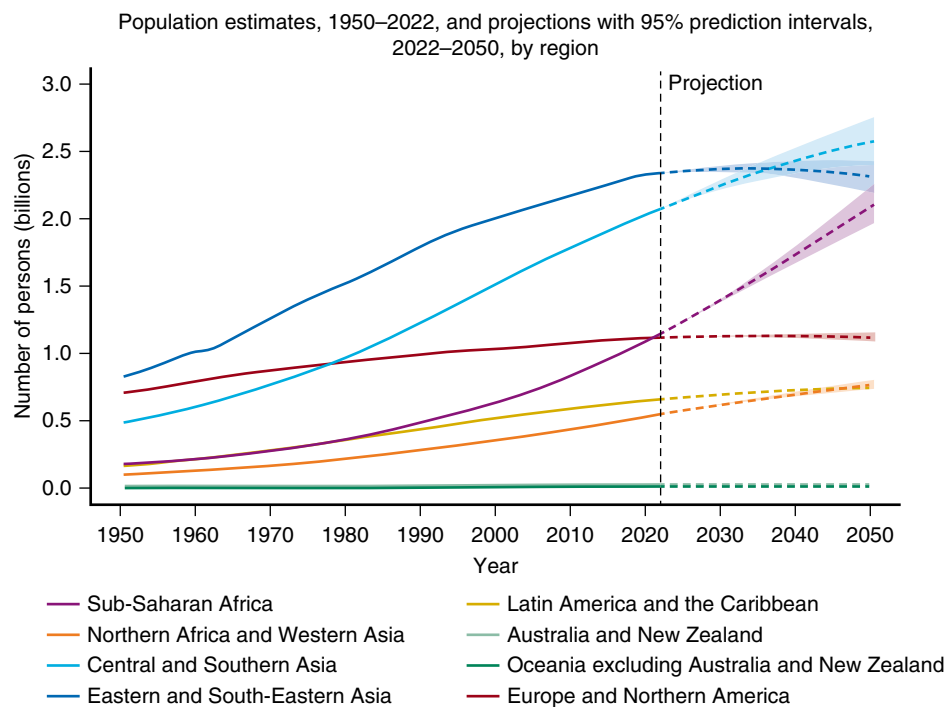


Fig. 75.1 World population estimates by region, with projections to 2050. (From World Population Prospects 2022. Summary of results. United Nations, Department of Economic and Social Affairs, Population Division; 2022.)

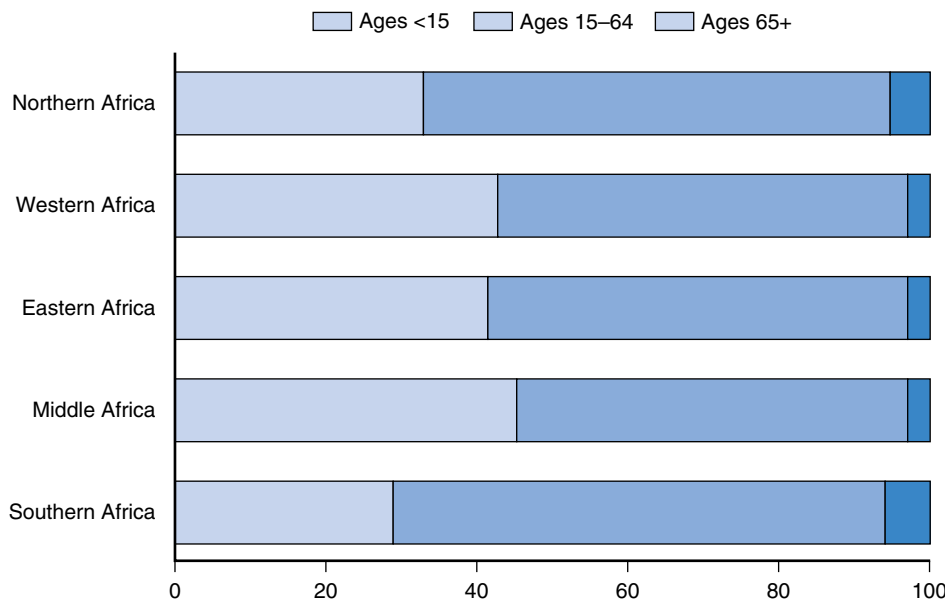


Fig. 75.2 Age distribution of the African population, by subregion. (From Africa Overview. 2022 World Population Data Sheet. Population Reference Bureau; 2022.)

lifestyles, NCDs are fast becoming the leading cause of death throughout Africa. In sub-Saharan Africa, NCDs were responsible for 37% of deaths in 2019, rising from 24% in 2000.⁶ In this region, the number of people living with diabetes is expected to reach 47 million by 2045, up from 19 million in 2019.

Four key risk behaviors for NCDs—tobacco and alcohol use, physical inactivity, and unhealthy diet—are on the rise among young Africans.⁷ Overweight and obesity,

especially among girls, are emerging as critical public health issues even where undernutrition still remains a problem.

Africans are using more tobacco and are starting to smoke at younger ages, with the 2013–2018 Global Youth Tobacco Survey indicating a prevalence of current tobacco product use among adolescents of 19%.⁸ Harmful alcohol consumption, a risk factor for NCDs and injuries, is expected to increase with further economic development.⁹

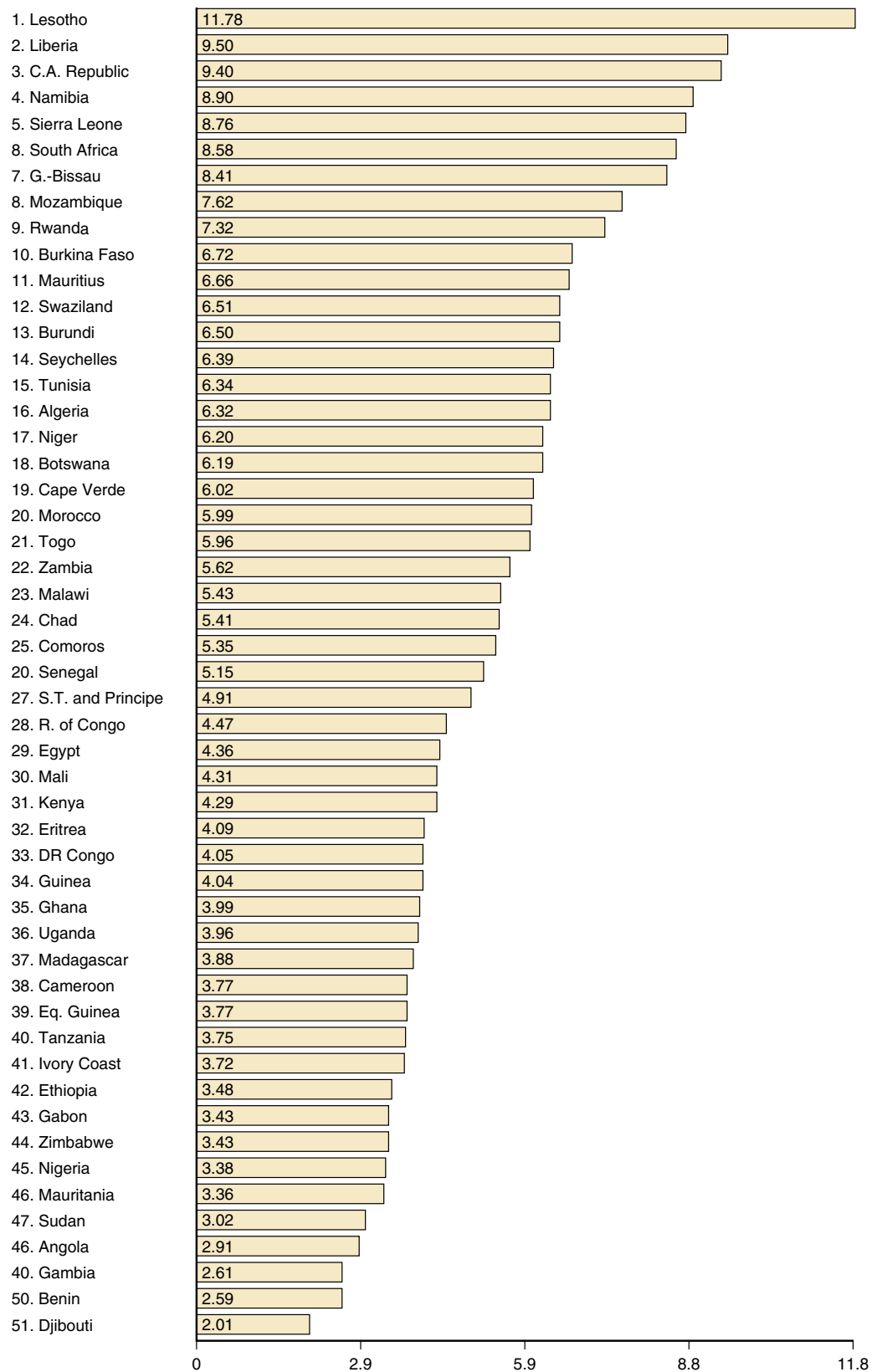


Fig. 75.3 Total expenditure on health as percentage of gross domestic product in the Africa region, 2020. (From The World Bank. Health spending as percent of GDP—country rankings. [TheGlobalEconomy.com](https://www.theglobaleconomy.com/rankings/health_spending_as_percent_of_gdp/Africa/), 2020. <https://www.theglobaleconomy.com/rankings/health_spending_as_percent_of_gdp/Africa/>; 2023. Accessed 13.11.23.)

Africa already has the highest prevalence of heavy episodic drinking of any region.

The burden from cancer is expected to more than double between 2008 and 2030, the drivers including HIV and other infections, tobacco use, occupational and environmental risks such as air pollution and exposure in mining, and aging of the population.

Injuries, such as those related to violence or road traffic accidents, also contribute a substantial disease burden in Africa. Sub-Saharan Africa has the highest road traffic death rate in the world, estimated at 24 people per 100,000 population per year.^{9,10} Conflicts are another major cause of death and disability.¹¹ While most of the deaths and injuries affect men, there are high rates of sexual violence against women in conflict situations, as documented for the Liberian civil war and Rwandan genocide. In conflict-affected countries, poverty levels are usually high and welfare levels low. The stability and social cohesion necessary for development are lacking, and there is often breakdown of death registration and other statistical monitoring systems.

Improvements in HIV/AIDS-related mortality rates and life expectancy have been recorded in many African countries, mainly as a result of the increasing availability of antiretroviral treatment (ART). In the Africa region in 2022, an estimated 25.6 million people were living with HIV, of whom 90% knew their status, 82% were receiving treatment, and 76% had suppressed viral loads.¹² Although declining, the number of new infections is still high, with an estimated 660,000 people acquiring HIV in 2022. Young women are the most vulnerable because of gender-based violence and less access to educational and economic opportunities.

The COVID-19 pandemic presented huge challenges to African countries but, fortunately, the growth in case numbers was not exponential, the disease severity was milder than seen elsewhere, and the anticipated overwhelming of the health services did not occur.¹³ Nevertheless, there were still many deaths from COVID-19 in patients requiring kidney replacement therapy (KRT), with mortality rates approximating 20%. Transplantation programs were suspended because of concerns regarding the consequences of SARS-CoV-2 infection in immunocompromised patients, and the management of potential deceased donors was stopped so that emergency departments and intensive care units could prioritize patients with COVID-19.¹⁴

HUMAN RESOURCES FOR NEPHROLOGY

According to the World Health Organization (WHO) Global Health Observatory, the Africa region has the lowest density of medical doctors (around 2 per 10,000 population), and nursing and midwifery personnel (around 10 per 10,000 population).¹⁵ Africa also has the lowest number of nephrologists per million population (pmp) in the world, with many countries having no nephrologists.

The latest report from the International Society of Nephrology (ISN) Global Kidney Health Atlas (ISN-GKHA)¹⁶ includes data from 41 of the 54 African countries, representing 93% of the region's population. The median density of nephrologists is 1.1 pmp, compared with the global median of 11.8 pmp. Women comprise 25% of the

nephrologists in Africa. The highest nephrologist densities (>15 pmp) were reported from Lesotho, Egypt, Tunisia, and Morocco, whereas many countries had a density of <1 pmp (Fig. 75.4). More than three-quarters of the participating countries reported shortages of professionals involved in kidney care: nephrologists (85%), pediatric nephrologists (98%), transplant surgeons (88%), surgeons for vascular access (88%) and peritoneal dialysis catheter placement (85%), vascular access coordinators (83%), dietitians (80%), and transplant coordinators (76%).¹⁶ Training initiatives should be expanded to include general practitioners and nurses who can implement effective interventions to reduce the burden of kidney disease in primary care settings¹⁷ and must also include the nurses and technologists who play a vital role in providing acute and chronic dialysis therapies.

Brain drain continues to pose a great threat to the limited nephrology health workforce on the African continent despite various recommendations¹⁸ and global initiatives.¹⁹ The reasons for physician emigration include poor working conditions and remuneration, limited career opportunities, low standards of living, and sociopolitical unrest.²⁰

Since 1985, the ISN Fellowship Program has sponsored the training of 223 Fellows from Africa. In the first 10 years of the program, all African Fellows trained in Western Europe or North America, whereas in the past 10 years nearly half trained in African centers (mostly in South Africa). A survey covering the period 2010–2019²¹ revealed an increased focus on clinical training, as well as an increase in the research impact, with 44% of respondents reporting publications and 27% reporting delivering congress presentations.

ACUTE KIDNEY INJURY

Epidemiologic data on AKI are scarce in Africa. The ISN's 0by25 Global Snapshot remains the most comprehensive study, conducted over a 10-week period in 2014 (see Chapters 27 and 28).²² The mean age of the African patients was 53 years. Community-acquired AKI accounted for 84% of the cases, and 19% were seen in intensive care unit settings.²³ The most common causes of AKI were dehydration, sepsis, hypotension, and infection (Fig. 75.5). Common infections associated with AKI in Africa include malaria, HIV, leptospirosis, diarrheal illnesses, hemorrhagic fevers, and tick-borne illnesses.²⁴ The epidemiology of AKI differs between North Africa and sub-Saharan Africa, although the cases are mostly community acquired in both regions, and infection and hypovolemia remain the principal causes. In sub-Saharan Africa,²⁵ infection was the leading cause in both adults and children and other common etiologies included nephrotoxins, pregnancy complications and glomerulonephritis in adults, and glomerulonephritis, nephrotoxins, hypovolemia, and urinary tract obstruction in children (Fig. 75.6). From Morocco, the IRAM Survey²⁶ included 541 cases of AKI. The main causes were volume depletion, sepsis, and obstruction of the urinary tract.

ACUTE KIDNEY INJURY AND MALARIA

According to the World Malaria Report, in 2021, 95% of all cases of malaria (234 million) and 96% of all malaria deaths (593,000) occurred in sub-Saharan Africa. Almost

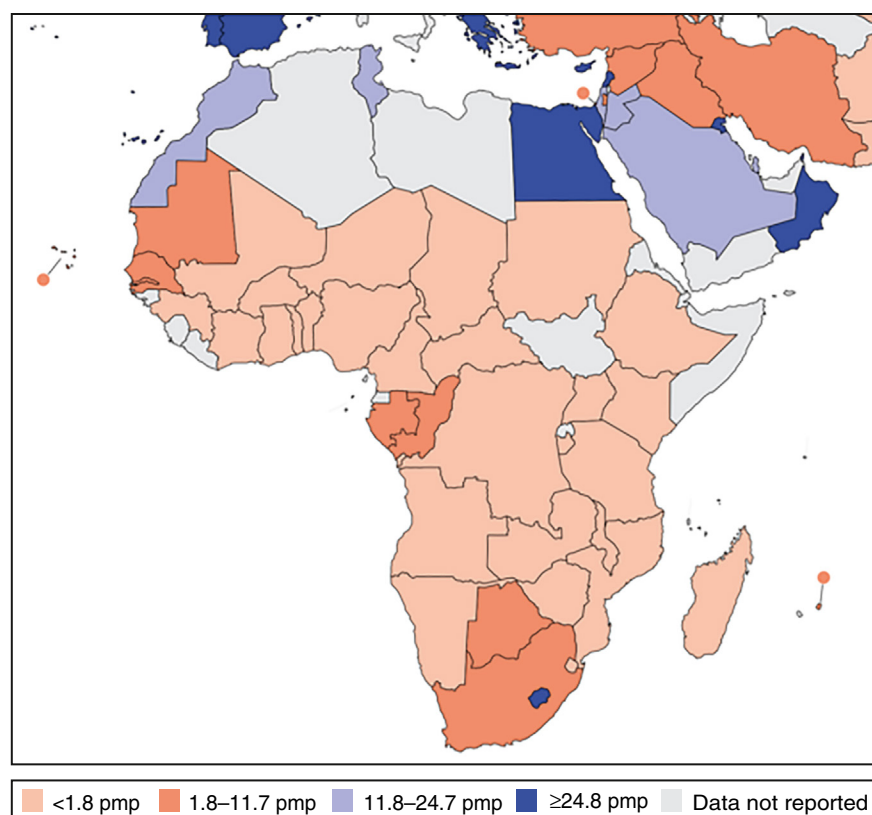


Fig. 75.4 Prevalence of nephrologists in African countries, per million population. (From Bello AK et al., 2023, ISN-Global Kidney Health Atlas, ISN, Brussels. Available at www.theisn.org/global-atlas.) Map lines delineate study areas and do not necessarily depict accepted national boundaries.

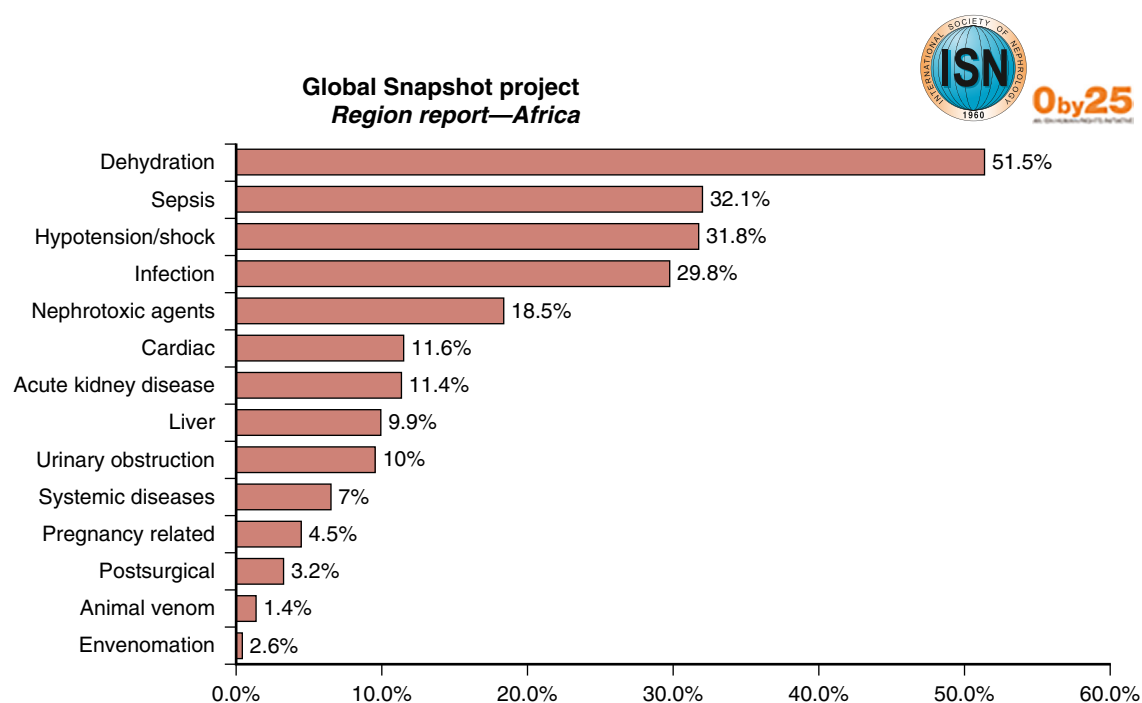


Fig. 75.5 Etiologic factors for acute kidney injury in Africa. (From Mehta RL, Macedo E, Zhang J, ISN Oby25 Initiative Steering Committee. International Society of Nephrology Oby25 Global Snapshot. Regional report for Africa; 2015.)

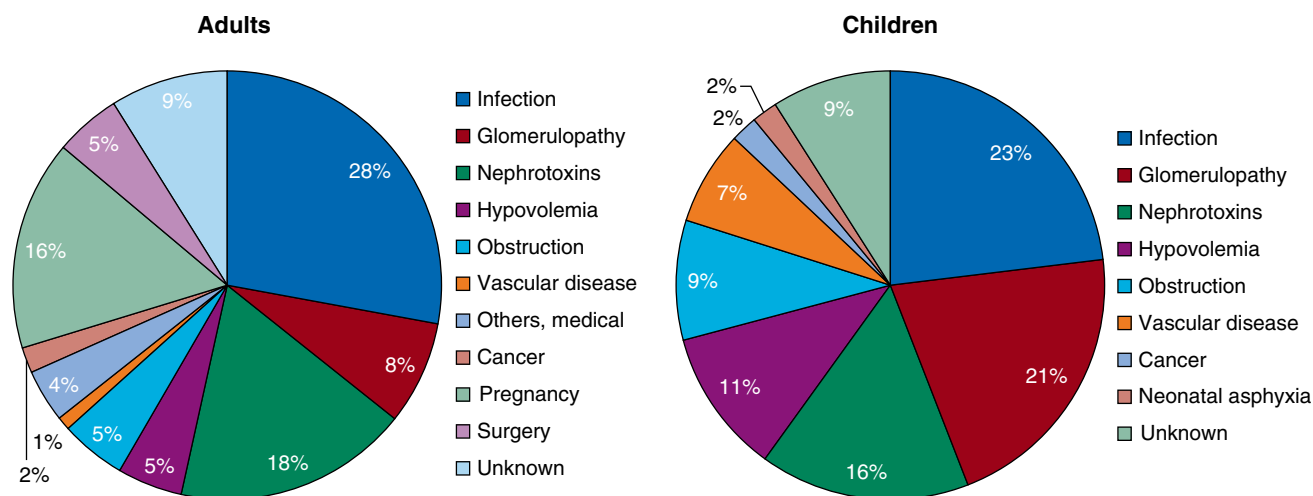


Fig. 75.6 Prevalence of etiologic factors for acute kidney injury in adults and children in sub-Saharan Africa. (Adapted from Olowu WA, Niang A, Osafo C, et al. Outcomes of acute kidney injury in children and adults in sub-Saharan Africa: a systematic review. *Lancet Glob Health*. 2016;4:e242–250.)

half of all global cases occurred in four countries: Nigeria, the Democratic Republic of the Congo, Uganda, and Mozambique. Almost 80% of all deaths occurred in children under the age of 5 years. In 2021, the estimated percentage of the population with access to an insecticide-treated mosquito net and the percentage of the population sleeping under these nets were 54% and 47%, respectively.²⁷ The report also documents a decline in the use of another important preventative measure, spraying the inside walls of homes with insecticides. The percentage of the African population protected by indoor spraying declined from 10.7% in 2010 to 5.3% in 2021. Access to seasonal malaria chemoprevention and the malaria vaccines are beginning to be rolled out, offering hope for more effective prevention in the future.

Malaria-associated AKI is usually due to *Plasmodium falciparum* infection. The incidence of AKI approaches 60% in patients with severe disease, with risk factors including severe parasitemia, children younger than the age of 5 years, pregnant women, and HIV infection.^{24,28} The mortality rate is approximately 45% and highest in children.²⁹ Early antimalarial treatment with artemisinin-based therapies, good fluid and electrolyte management, and dialysis, when indicated, is associated with improved patient survival and recovery of kidney function in patients with malaria-induced AKI (see [Chapter 59](#)).

ACUTE KIDNEY INJURY IN HUMAN IMMUNODEFICIENCY VIRUS INFECTION

AKI is a common complication in HIV-infected patients compared with uninfected patients, with an odds ratio of 2.8 for those treated with ART, and 4.6 in the pre-HAART era.³⁰ Patients hospitalized with complications of HIV may be at increased risk of AKI that develops in relation to volume depletion, hemodynamic stress, infections, malignancy, and administration of nephrotoxic medication or radiocontrast material.

Kidney injury related to ART occurs in <10% of patients. Tenofovir disoproxil fumarate (TDF) may cause mitochondrial toxicity in proximal renal tubular cells. This may result in AKI, CKD, or Fanconi syndrome. While TDF is well tolerated by most patients, there is increased

risk of TDF-related renal injury in those with preexisting kidney impairment, older age, low body weight, advanced HIV disease, comorbidities (e.g., diabetes, hypertension, and hepatitis C coinfection), concomitant use of other nephrotoxic drugs and protease inhibitors.³¹

OBSTETRIC ACUTE KIDNEY INJURY

Obstetric AKI continues to be a frequent clinical problem in Africa, mainly secondary to preeclampsia and eclampsia, sepsis, and obstetric hemorrhage.³² Recent systematic reviews of studies from Africa have reported an overall prevalence of 6%, without any difference by subregion, and exceptionally high perinatal mortality rates ranging from 15% to 60%.^{32–34}

ACUTE KIDNEY INJURY AND TOXINS

The use of traditional medicines is common in low- and middle-income countries (LMICs), where its use is related to spiritual beliefs, the need for protection or cleansing, sexual potency or fertility, suspicion of conventional medicine, and poor access to medical care.³⁵ In sub-Saharan Africa, more than 80% of people use traditional medicines as their principle form of health care. The medicines are used for numerous conditions including diabetes, hypertension, and CKD, in both urban and rural settings and by people of all socioeconomic classes.³⁶

Traditional medicines are mostly used without major adverse effects; however, their use may delay patients' presentation to medical care, and many have been associated with kidney injury ([Table 75.1](#)).³⁶ Volume depletion is a major risk factor for the development of AKI associated with traditional medicines. In LMICs, such remedies account for up to 35% of AKI cases and the reported mortality rates range from 24% to 75%.³⁵ In those who survive, recovery of kidney function is often incomplete, resulting in CKD. As the use of traditional medicines is generally underreported to physicians, it should always be considered in the differential diagnosis of any community-acquired or unexplained kidney injury.³⁷

Table 75.1 Examples of Traditional Medicines Used Across Sub-Saharan Africa that Have Known Nephrotoxic Effects

Nomenclature		Common Uses or Indications	Nephrotoxic Effects
Scientific Name	Common English Name		
<i>Aloe vera</i> species inducing ferox and secundiflora	Cape aloes Aloe vera	Southern Africa: arthritis, burns/skin conditions, hypertension, purging/laxative, dyspepsia, antiinflammatory, cosmetics, eye ailments, conjunctivitis, venereal diseases, infertility, impotence East Africa (Kenya, Uganda, Ethiopia, and Tanzania): malaria, purging/laxative for cleansing purposes, dyspepsia, skin ulcerations/wound healing, cosmetic, infertility, antiparasitic	Volume depletion and electrolyte imbalance Acute tubular necrosis Acute interstitial nephritis
<i>Euphorbia matabalensis</i>	Three-forked shrub (Euphorbia)	Eastern Africa (including Somalia, Kenya, Tanzania, Malawi, Mozambique, and Zimbabwe): constipation, purging, hypertension Southern Africa (including Zambia, Botswana, Angola, and Namibia): abortifacient, constipation, purging, hypertension, venereal diseases, dyspepsia, myalgias	Acute tubular necrosis: directly nephrotoxic, perhaps attributable to latex compounds
<i>Cymbopogon citratus</i>	Lemongrass	South Africa: diabetes, oral thrush Nigeria: fever/malaria, stimulant, antispasmodic Cameroon: malaria, jaundice Angola: cough, antiemetic, antiseptic, arthritis/arthritis	Volume depletion, diarrhea Chronic decrease in glomerular filtration rate, chronic interstitial nephritis
<i>Securidaca longepedunculata</i>	African violet tree Wild wisteria	Zambia: abortifacient Burkina Faso: malaria, skin disorders Nigeria: abortifacient, constipation, fever, aphrodisiac Botswana and Eastern Africa (including Tanzania, Malawi, Uganda, Kenya, and Zimbabwe): epilepsy, arthritis, purging, constipation, wound healing, snakebites, fever/malaria, impotence, dysmenorrhea	Acute tubular necrosis: causes cortical necrosis, volume depletion, and contains salicylates thought to contribute to renal vasoconstriction Acute interstitial nephritis
<i>Crotalaria</i>	Wild sunhemp	South Africa, Botswana and Eastern Africa	Acute tubular necrosis: contains nephrotoxic alkaloids
<i>Laburnifolia</i>	Rattlepod birdflower	(including Mozambique and Zimbabwe): enema, purging, dysmenorrhea, abortifacient, dyspepsia, antispasmodic	Hepatorenal failure
<i>Callilepis laureola</i>	Ox-eye daisy Impila	Mozambique and South Africa: dyspepsia, antiparasitic, impotence/infertility, purging, evil spirits	Acute tubular necrosis and acute interstitial nephritis Hepatorenal failure and chronic decrease in glomerular filtration rate Hyperkalemia

From Stanifer JW, Kilonzo K, Wang D, et al. Traditional medicines and kidney disease in low-and middle-income countries: opportunities and challenges. *Sem Nephrol*. 2017;37:245–259.

The nephrotoxicity of a remedy may be related to nephrotoxic components but also to contamination of the remedy, the erroneous identification or preparation of plants, incorrect use, interaction with other medications, and patient factors such as comorbid conditions and age.³⁵ Other nephrotoxic agents encountered in some African countries, especially in East Africa and the Maghreb, include naphthalene,³⁸ mercury in skin-lightening creams,^{39–41}

and hair dye and henna preparations that contain paraphenylenediamine.⁴²

Animal toxins are also a major cause of AKI in Africa. Snake venom may cause neurotoxicity, cardiotoxicity and hematologic toxicity, as well as tissue necrosis.⁴³ Bites occur especially in plantations and commonly involve *Causus maculatus* (spotted night adder), *Naja melanoleuca* (black cobras), and *Dendroaspis* species (green mambas). In

forested areas, gaboon vipers (*Bitis* species) are common while in savanna areas *Echis* species are common and cause the greatest numbers of deaths by envenomation in Africa.⁴⁴ The range of kidney pathology includes acute tubular necrosis, cortical necrosis, acute interstitial nephritis, and proliferative glomerulonephritis. Scorpion stings are a significant public health problem in north-Saharan Africa, Sahelian Africa, and South Africa.⁴⁵ The scorpion α -toxins cause neuronal excitation,⁴⁶ and AKI may develop as a result of cardiac dysfunction, disseminated intravascular coagulation, and hemorrhage.

THERAPEUTIC CHALLENGES IN ACUTE KIDNEY INJURY

In most African countries, limited resources and health infrastructure are associated with late presentation and poor recognition and treatment of AKI.²⁵ In studies reporting dialysis need, 66% of children and 70% of adults had indications for dialysis but only 64% and 33% of these patients received dialysis, respectively. The mortality was 73% in children and 86% in adults when dialysis was needed but not received. Major barriers to access to care were unaffordable out-of-pocket costs, limited hospital resources, late presentation, and female sex.²⁵ In another review of AKI in Nigerian children between 1990 and 2012, dialysis access rates ranged between 15% and 68% of those with an indication for dialysis.⁴⁷ In North Africa, dialysis for AKI is more easily accessible in hospital settings. In a Moroccan survey, dialysis was needed in 40% of cases and intermittent hemodialysis was the predominant treatment modality. In-hospital mortality was 25%, dialysis dependence was 7%, residual kidney failure was present in 39%, and complete recovery was documented at 29%.²⁶

Two global initiatives have impacted the outcomes of patients with AKI, especially in low-resource settings. In 2013, the ISN launched “0by25,” an ambitious initiative to eliminate preventable deaths from AKI by 2025.⁴⁸ The project focuses on evidence and awareness, as well as on training for health workers to increase the early detection of AKI and appropriate use of simple management, such as fluid repletion. The second initiative is “Saving Young Lives,” which aims to develop acute peritoneal dialysis programs in low-resource settings.^{49,50} Peritoneal dialysis can be lifesaving in treating AKI where hemodialysis is not available. Locally prepared peritoneal dialysis fluid adapted from commonly available intravenous solutions can be used safely and effectively.⁵¹⁻⁵³ Practical hands-on training covering all aspects of acute peritoneal dialysis has already been provided to 69 nurses and 209 doctors at an annual training course held at the Red Cross War Memorial Children’s Hospital in Cape Town, South Africa.

Importantly, a large proportion of AKI cases and their clinical consequences are potentially preventable.⁵⁴ Education regarding the avoidance of nephrotoxins, prompt treatment of infections, and fluid replacement are important facets of therapy,⁵⁵ especially in countries with limited access to kidney replacement therapies.

LONG-TERM OUTCOMES OF ACUTE KIDNEY INJURY IN AFRICA

In Africa, AKI may be an important contributor to the burden of CKD. Two prospective studies from Cape Town, South

Africa, have reported on outcomes at 3 months. Dlamini and colleagues⁵⁶ studied 366 patients with AKI and reported that 39% had died within 3 months. Among the survivors, 119 had follow-up creatinine measurements and of these, 80% had completely recovered kidney function and 3.4% had developed end-stage kidney failure (ESKF). Chothia and colleagues⁵⁷ reported on the 3-month outcomes of 113 patients and reported a composite outcome of CKD, ESKF, or death in 57% of cases, with rates of 10%, 3%, and 44%, respectively. From Côte d’Ivoire, Yao and colleagues⁵⁸ reported a mortality rate of 37.6% in patients with AKI and the development of CKD in 42% of the survivors. Older age, hypertension, anemia, severe AKI, malignant hypertension, and cancer were associated with abnormal kidney function beyond 3 months.

CHRONIC KIDNEY DISEASE

CKD is a major public health problem, with an estimated global prevalence of between 11% and 13%.⁵⁹ The drivers of CKD in Africa differ from what has been described globally. Apart from traditional risk factors such as diabetes, hypertension, and aging, there are unique risk factors such as chronic or repeated acute infections, trauma, heat stress, envenomation, and pregnancy-related complications, which may cause AKI, acute kidney disease, and, ultimately, CKD (Fig. 75.7).⁶⁰

People from disadvantaged communities are at particularly high risk of unrecognized and untreated CKD.⁶¹ This includes indigenous communities and racial and ethnic minorities.⁶² Low socioeconomic status affects the incidence and progression of CKD in LMICs through poor education, an inadequate nephrology workforce, weak health systems, and poor health financing.⁶³

ISSUES IN THE DIAGNOSIS, SCREENING, AND PREVENTION OF CHRONIC KIDNEY DISEASE

The accurate diagnosis and staging of CKD is especially important in Africa as options for treating late complications such as kidney failure are limited.⁶⁴ Screening those at higher risk of developing CKD is essential. The choice of methods used to assess GFR can have a major impact on determining if an individual has CKD or not, on the stage of CKD, and on the reported population prevalence. In Africa, GFR is almost exclusively estimated due to a lack of access to methods used to measure it. Holness and colleagues⁶⁵ have suggested that measuring GFR by iothexol clearance on dried blood spots may offer a solution in the African context.

Studies in African populations have found that the currently recommended GFR prediction equations overestimate kidney function, with consequent underestimation of the prevalence of CKD.⁶⁶ Eight African studies, five from South Africa,⁶⁷⁻⁷¹ one from Côte d’Ivoire,⁷² one from Kenya,⁷³ and one multinational study that included South Africa, Malawi, and Uganda⁷⁴ have compared prediction equations with GFR measured by gold standard methods. All studies reported that the equations performed better without the “African American” ethnicity factor. Creatinine-based equations overestimated GFR; the use of cystatin C, alone or in combination with creatinine, resulted in less bias.⁷⁵

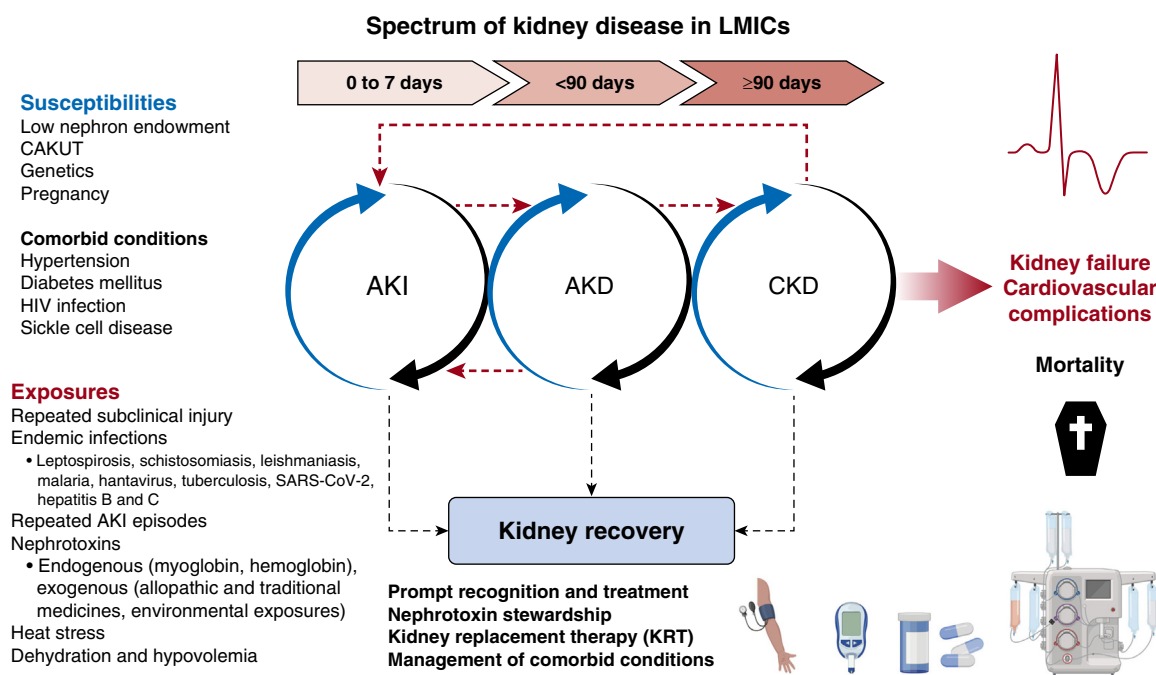


Fig. 75.7 The spectrum of kidney disease in low- and middle-income countries (LMICs). Acute kidney injury (AKI), acute kidney disease (AKD), and chronic kidney disease (CKD) are interconnected syndromes. AKD and CKD can occur from the progression of AKI but also increase susceptibility to repeated episodes of AKI. Populations in LMICs may be more susceptible to kidney disease through genetic risk factors, lower nephron endowment, and chronic exposures or repeated episodes of kidney injury across the lifespan. (Adapted with permission from Kalyesubula R, Conroy AL, Calice-Silva V, et al. Screening for kidney disease in low-and middle-income countries. *Semin Nephrol.* 2022;151315.)

Fabian and colleagues⁷⁶ have published a systematic review of the methods of measuring, estimating, and reporting kidney function in studies from sub-Saharan Africa. They highlight potential sources of bias and propose a checklist to standardize the reporting of kidney function and improve the quality of CKD studies from the region.

Many cross-sectional studies on CKD suffer from a common weakness in that they do not include repeat estimates of GFR or protein excretion and hence do not meet the criterion of chronicity for diagnosing CKD. When repeat measurements are done, this often results in lower estimates for CKD prevalence.^{77,78} In the Maremar study in Morocco,⁷⁹ investigators demonstrated that repeating eGFR after 3 months avoided a false-positive diagnosis of CKD in 32% of cases. Repeating dipstick testing after 1 to 2 weeks allowed positive results to be identified as false positives in 68% of subjects with mild proteinuria (dipstick +) and in 29% of those with overt proteinuria (dipstick ++ to +++).⁷⁹ Another African study of CKD prevalence, in sugarcane workers in Cameroon, also included the element of chronicity in their methods and reported a relatively low CKD rate of 3.4%.⁸⁰

Another challenge is the feasibility and cost-effectiveness of CKD screening and prevention programs in African countries. Diseases such as diabetes, hypertension, and HIV infection are the major causes of CKD in many countries,^{77,81,82} and it seems reasonable to couple CKD screening and preventive measures with already-implemented programs for diabetes, hypertension, and HIV infection.⁸³ For example, the WHO STEPS method is a simple, standardized method for collecting data on NCDs and their risk factors and could easily be adapted to include screening for CKD.⁸⁴ Kalyesubula and colleagues⁶⁰ have

proposed a modified screening algorithm (Fig. 75.8) for LMICs, where factors such as age >45 years, environmental exposures, and repeated or chronic exposure to infections may identify people at higher risk of CKD.

CHRONIC KIDNEY DISEASE PREVALENCE IN AFRICAN COUNTRIES

Differences in study sample sizes, CKD definition, lack of longitudinal testing, and heterogeneity in GFR estimation have been identified as contributing to the variable prevalence of CKD in Africa.⁸⁵

The first systematic review on the population prevalence of CKD in sub-Saharan Africa was published in 2014 by Stanifer and colleagues.⁸¹ The overall prevalence was 13.9%. Subsequent reviews by Kaze and colleagues⁸⁶ and Abd ElHafeez and colleagues⁸⁷ reported population prevalences of 15.8% and 10.1%, respectively. In both studies, the prevalence was highest in Western Africa and lowest in Northern Africa. Subsequent to the publication of these reviews, population prevalence studies have been published from several African countries, with estimates ranging from 6.4% to 16.1% (Table 75.2).

The prevalence of kidney disease among African children has been reviewed by Kayange and colleagues.⁸⁸ In 32 studies from nine sub-Saharan countries, the mean prevalence of proteinuria was 32.5%, hematuria 31.1%, and imaging abnormalities 14.8%. The quality of the studies was graded as moderate to poor. Few were recent, and none included the calculation of eGFR. Most were performed in areas where schistosomiasis is endemic, and this was the main risk factor identified.

Conditions associated with increased CKD risk in LMICs

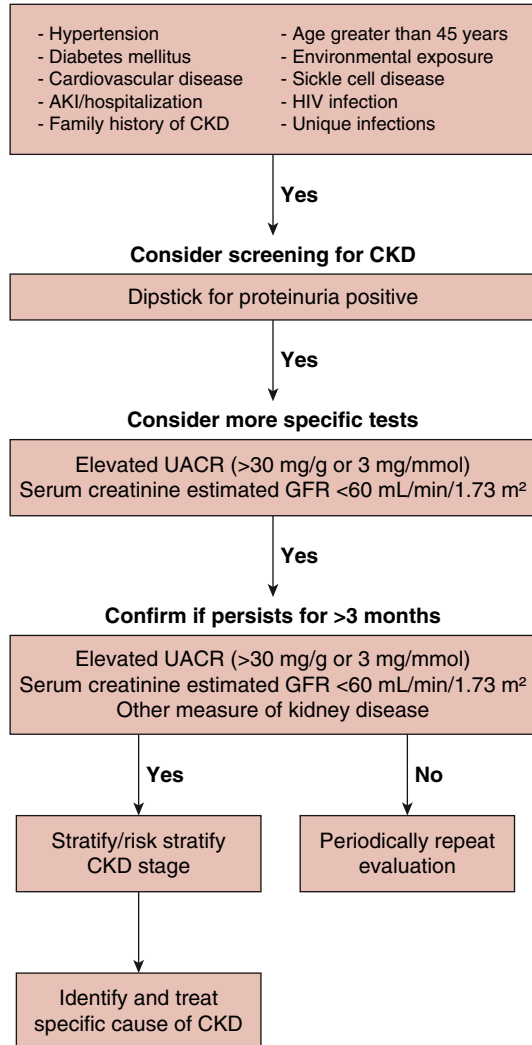


Fig. 75.8 Proposed screening algorithm for chronic kidney disease (CKD) in low- and middle-income countries (LMICs). AKI, Acute kidney injury; GFR, glomerular filtration rate; HIV, human immunodeficiency virus; UACR, urine albumin-to-creatinine ratio. (Adapted with permission from Kalyesubula R, Conroy AL, Calice-Silva V, et al. Screening for kidney disease in low- and middle-income countries. *Semin Nephrol.* 2022;151315.)

COMMON CAUSES OF CHRONIC KIDNEY DISEASE AND KIDNEY FAILURE

In many cases where CKD or kidney failure is documented in African patients, a specific diagnosis is never established. This may be because kidney biopsy services are not available or because patients present late, with advanced CKD, hypertension, and shrunken kidneys. A presumptive diagnosis of “hypertensive kidney disease” or “chronic glomerulonephritis” is often made, without good grounds for making such a diagnosis.⁸⁹ It has been recommended that the label “CKD—cause uncertain/unknown” be used in such cases and whenever there is uncertainty about the primary kidney disease.

Data on the etiology of kidney failure are available from renal registries for some African countries (Table 75.3).

Table 75.2 Recent Population Prevalence Studies from African Countries

Author and Year	Country	Prevalence
Gbaguidi et al., 2021 ²²⁷	Benin	16.1%
George et al., 2019 ²²⁸	Burkina Faso, Ghana, Kenya, South Africa	10.1%
Kaze et al. ²²⁹	Cameroon	11.7%
Masimango et al., 2020 ²³⁰	Democratic Republic of the Congo	12.2%
Olanrewaju et al., 2020 ²³¹	Nigeria	12.0%
Wokoma et al., 2017 ²³²	Nigeria	11.7%
Adeniyi et al., 2017 ²³³	South Africa	6.4%
Hodel et al., 2018 ²³⁴	Tanzania	13.6%
Plath et al., 2018 ²³⁵	Tanzania	12.4%
Muiru et al., 2020 ²³⁶	Uganda and Kenya	6.8%

Hypertension is reported as the most common primary kidney disease, and a large proportion of patients have no clearly identifiable cause for their kidney failure.

DIABETIC NEPHROPATHY

The Africa region is projected to experience the highest growth rates in the number of people with diabetes, with the total number increasing by 129% to 55 million by 2045.⁹⁰ These increases are driven by sedentary lifestyles, unhealthy diets, obesity, population growth, and aging.

Noubiap and colleagues⁹¹ included 32 studies in a systematic review of diabetic nephropathy in Africa. The reported prevalence varied from 11% to 83.7%. The rate of ESKF at 5 years of follow-up was 34.7%, and the mortality rate from diabetic nephropathy was 18.4% at 20 years of follow-up. More recently, Wagnaw and colleagues⁹² performed a meta-analysis to estimate the prevalence of diabetic nephropathy in sub-Saharan Africa. Most of the 27 included studies used proteinuria as evidence of nephropathy, and one-third were conducted in patients with type 2 diabetes mellitus. The pooled prevalence of diabetic nephropathy was 35.3%. The prevalence was highest in patients with type 2 diabetes mellitus (41.4%) and in the Southern African region (40.4%).

HYPERTENSIVE KIDNEY DISEASE

A systematic review by Ajayi and colleagues⁹³ assessed the prevalence of CKD from studies in Africa reporting target organ damage related to hypertension. The review included 12 studies (5297 participants) from six African countries: Ghana, Nigeria, Uganda, Tanzania, the Democratic Republic of the Congo, and South Africa. The overall prevalence of CKD among hypertensive patients was 18%, with the highest prevalence observed in West Africa (21%) and urban areas (20%). In registry reports from South Africa,⁹⁴ Ghana,⁹⁵ and Egypt,⁹⁶ hypertension was reported as the etiology of kidney failure in 34% to 41% of patients on KRT (Table 75.2). Unpublished data from Kenya⁹⁷ and Senegal⁹⁸ also

Table 75.3 Etiology of Kidney Disease in Patients on Kidney Replacement Therapy (% of Total) Reported in Recent Registry Reports from African Countries

Study	Country	Last Data	Cause Unknown	Hypertension	Diabetes	GN	Cystic Disease
Hassaballa et al., 2022 ⁹⁶	Egypt	2020	13.4	41.3	13.6	3.2	4.9
Boima et al., 2021 ⁹⁵	Ghana	2017	28.5	37.8	9.2	7.7	0.7
Davids et al., 2023 ⁹⁴	South Africa	2021	34.1	33.7	14.4	9.5	2.9

highlight hypertensive kidney disease as a leading cause of kidney failure.

It is generally accepted that accelerated and malignant forms of hypertension can lead to kidney failure.^{99,100} although recovery of kidney function sometimes occurs months after starting KRT, especially when patients are treated with peritoneal dialysis.¹⁰¹ In a study published in 1982, Gold and colleagues¹⁰² reviewed the renal histology in 65 Black South African patients on chronic hemodialysis and reported that primary malignant hypertension was the most common cause, occurring in 49%. Large South African biopsy series have been published more recently, reporting hypertensive kidney disease in only 1.3% (Cape Town, 2227 biopsies)¹⁰³ and 6.2% (Johannesburg, 1848 biopsies)¹⁰⁴ of their biopsies, respectively.

Nephrologists also often apply a diagnosis of “hypertensive kidney disease” or “hypertensive nephrosclerosis” to nondiabetic patients, especially Black patients, who present with CKD and only mild-to-moderate hypertension.⁸⁹ It now seems clear that many of these patients have glomerular disease, which is related to the presence of renal-risk variants of the APOL1 gene, and have secondarily elevated blood pressure.¹⁰⁵ For example, in the African American Study of Kidney Disease and Hypertension, the APOL1 association was strongest in individuals with more proteinuria and progressive nephropathy. Aggressive hypertension control and the use of angiotensin-converting enzyme inhibitors failed to halt progression, supporting the role of mechanisms other than hypertension.¹⁰⁶ The American Study of Kidney Disease and Hypertension patients predominantly had focal glomerulosclerosis with interstitial and vascular changes that did not correlate with blood pressure.¹⁰⁷ This information prompted calls for the term “hypertensive nephrosclerosis” to be abandoned.^{89,108}

The South African Renal Registry recommends that its users submitting data indicate hypertensive kidney disease as the primary diagnosis only if there is no reason to suspect that the hypertension may be secondary to preexisting kidney disease,⁹⁴ according to the following criteria: hypertension known to precede kidney dysfunction, left ventricular hypertrophy, proteinuria <2 g/day and no evidence of other kidney diseases.^{109,110}

GLOMERULONEPHRITIS

Okpechi and colleagues¹¹¹ conducted a systematic review of histologically proven glomerulonephritis in Africa between 1980 and 2014. They noted declining kidney biopsy rates

and the consequent paucity of data on pathologic patterns of kidney diseases from Africa. Of the 24 studies (12,093 biopsies) from 13 countries, 70% originated from North African countries. Nephrotic syndrome was the most common indication for kidney biopsy. Minimal change disease represented 16.5%, focal segmental glomerulosclerosis (FSGS) 15.9%, mesangiocapillary glomerulonephritis (MCGN) 11.8%, membranous nephropathy 6.6%, IgA nephropathy 2.8%, and crescentic glomerulonephritis 2.0% of the primary kidney diseases. Nephritis related to hepatitis B (8.4%) and systemic lupus erythematosus (7.7%) had the highest prevalence among secondary glomerular diseases. Fig. 75.9 illustrates some of the differences between North African and sub-Saharan African countries.

Esmail and colleagues¹⁰³ reviewed 2227 native kidney biopsies performed at a tertiary center in Cape Town, South Africa, over 22 years. The most common primary glomerular disease was MCGN (34.5%), followed by FSGS (22.3%) and membranous nephropathy (15.8%). Lupus nephritis was the most common secondary glomerular disease (39.1%), followed by HIV-associated nephropathy (HIVAN) (22.1%). Vermeulen and colleagues¹⁰⁴ published a series of 1848 native kidney biopsies performed more than 30 years in Johannesburg, South Africa, and reported FSGS as the most common primary glomerular disease (29.6%), followed by membranous nephropathy (25.7%) and MCGN (18.1%). Lupus nephritis (31%) and HIVAN (13.3%) were also the most common secondary glomerular diseases identified.

Following the introduction of the new classification for MCGN, there has been a dearth of data on C3 glomerulonephritis from Africa. A recent study by Hargey and colleagues¹¹² reported on the outcomes of 19 patients with C3 glomerulonephritis in Cape Town, South Africa. During a median follow-up period of 15 months, nearly 80% of the patients had experienced the composite outcome of CKD, ESKF, or death.

Uwaezuoke and colleagues¹¹³ conducted a systematic review to investigate the most common histopathologic diagnoses in steroid-resistant childhood nephrotic syndrome in sub-Saharan Africa. Four of the included studies were from Nigeria, and two from South Africa. The most common diagnoses were FSGS (57.2%), MCGN (22.4%), minimal change disease (14%), and membranous nephropathy (6.7%).

Glomerular diseases also common in Africa include postinfectious glomerulonephritis and those associated with lupus, hepatitis B, and schistosomiasis. These conditions are discussed in detail elsewhere in this book.

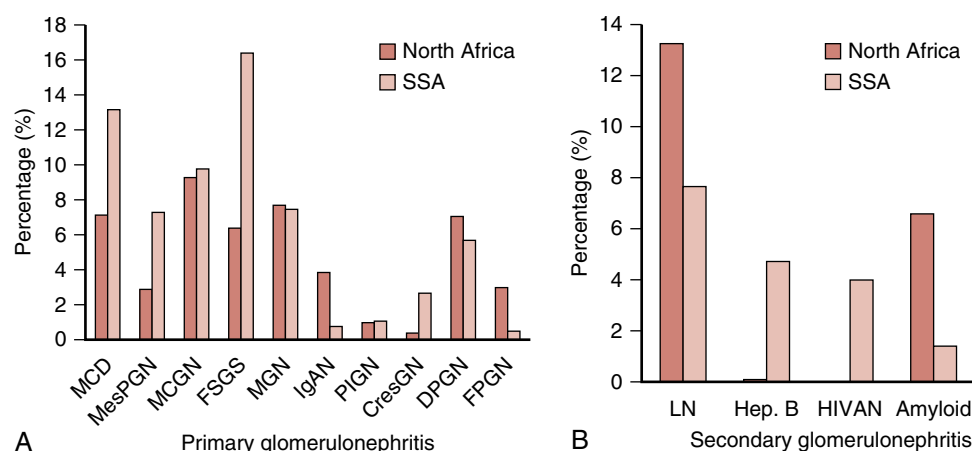


Fig. 75.9 Differences between North Africa and sub-Saharan Africa in the prevalence of common glomerular diseases. *CresGN*, Crescentic glomerulonephritis; *DPGN*, diffuse proliferative glomerulonephritis; *FPGN*, focal proliferative glomerulonephritis; *FSGS*, focal segmental glomerulosclerosis; *Hep B*, hepatitis B virus–associated nephropathy; *HIVAN*, HIV-associated nephropathy; *IgAN*, IgA nephropathy; *LN*, lupus nephritis; *MCD*, minimal change disease; *MCGN*, mesangiocapillary glomerulonephritis; *MesPGN*, mesangioproliferative glomerulonephritis; *MGN*, membranous glomerulonephritis; *PIGN*, postinfectious glomerulonephritis; *SSA*, sub-Saharan Africa.

KIDNEY DISEASE ASSOCIATED WITH HIV INFECTION

According to the 2022 UNAIDS Global AIDS Update,¹¹⁴ progress toward ending the AIDS pandemic is slowing rather than accelerating. The COVID-19 pandemic and economic and humanitarian crises have placed the global HIV response under increasing threat.

Eastern and Southern Africa remain the regions of the world with the heaviest burden of infection. In 2022, 20.8 million people in these regions were living with HIV, there were 500,000 new HIV infections, and there were 260,000 HIV-related deaths.¹¹⁵ The proportion of patients who were on treatment was 83%. Some African countries have already achieved the 95–95–95 targets (95% of people living with HIV know their HIV status, 95% of those receive sustained ART, and 95% of people receiving ART have viral suppression).¹¹⁶ These include Botswana, Eswatini, Rwanda, Tanzania, and Zimbabwe. An additional eight sub-Saharan African countries are also close to doing so.

A broad spectrum of kidney diseases is seen in patients with HIV infection. This includes diseases directly related to the HIV infection, those related to superinfections, those related to the immune response, and those related to the treatment. HIVAN is the classic lesion described, presenting with nephrotic-range proteinuria and decreased renal function but, because of renal salt wasting, patients often do not have hypertension or significant edema. Renal histology reveals a collapsing form of FSGS, with associated cystic tubular dilatation, interstitial inflammation, and the presence of tubuloreticular inclusion bodies. HIV RNA has been demonstrated in podocytes and tubular epithelial cells.¹¹⁷

HIVAN was first reported in 1984,^{118–120} with the first reports from Africa published in the late 1980s and early 1990s.^{121–123} The worldwide prevalence is highest in sub-Saharan Africa,¹²⁴ occurring especially in the absence of ART and in patients with markedly reduced CD4 counts and elevated viral loads.¹²⁵

Individuals of African descent have a genetic susceptibility to the development of HIVAN that is related to single nucleotide polymorphisms in the apolipoprotein L1 (*APOL1*) gene.¹²⁶ The G1 variant includes two missense mutations (*S342G*, *R573885319* and *I384M*, *R560910145*) that are in close linkage disequilibrium, and the G2 variant is a two-base pair deletion (*N388del*, *Y389del*, *R571785313*). These variants confer risk for HIVAN and HIV-associated FSGS, as well as other glomerular diseases and hypertensive nephrosclerosis. Kasembeli and colleagues¹²⁷ have demonstrated that HIV-positive, ART-naïve Black South Africans with two *APOL1* risk alleles are at high risk for developing HIVAN. In their study, 79% of patients with HIVAN and only 2% of controls carried two risk alleles. Individuals carrying any combination of two risk alleles had 89-fold higher odds of developing HIVAN compared with HIV-positive controls. It is estimated that 50% of HIV-positive individuals who have two *APOL1* risk alleles and do not receive effective ART will develop HIVAN.¹²⁸

There is wide variability in the prevalence of HIVAN in different African populations. This is likely to be related to the frequency of *APOL1* risk alleles. The highest frequency of the G1 and G2 variants is found in West Africa, while populations in North and East Africa have a much lower frequency of the risk alleles and of HIV-related kidney disease.¹²⁹ For example, in Kenya, Koech and colleagues¹³⁰ performed kidney biopsies on ART-naïve, HIV-positive adults who had persistent proteinuria. None of the biopsies revealed HIVAN; acute interstitial nephritis was the most common histologic diagnosis in this group of patients.

Kidney biopsy data on African patients with HIV infection have mostly come from South African studies. In Durban, Han and colleagues¹³¹ biopsied 30 patients with proteinuria who were not on ART and found that 83% had HIVAN. Gerntholtz and colleagues¹³² described the findings in 99 patients from the Johannesburg area; 27% of the biopsies showed classic HIVAN, 21% had HIV-related immune complex disease (HIV-ICD), and more

than half had diagnoses that were not directly related to the HIV infection. Also from Johannesburg, the biopsy series of Vermeulen and colleagues¹⁰⁴ included 364 HIV-infected patients; HIVAN was present in 32.7% and HIV-ICD in 11.8% of these patients. Wearne and colleagues¹³³ reviewed the biopsies of 192 HIV-infected patients in Cape Town, reporting that HIVAN was the most common histologic diagnosis (57%), with the collapsing subtype of FSGS present in 43%. Immune complex disease was present in 8.3% and a combination of HIVAN and HIV-ICD in 21.9%. The biopsy series of Esmail and colleagues,¹⁰³ from Cape Town, included 504 HIV-infected patients. The most common pathology in these patients was HIVAN (44.2%), followed by acute tubular necrosis (40.1%). Many patients had more than one histopathologic diagnosis.

Wearne and colleagues¹³⁴ (in 671 biopsies) and Diana and colleagues¹³⁵ (690 biopsies) have described a shift in the spectrum of kidney disease among HIV-infected South Africans since the introduction of ART in 2004. Tubulointerstitial disease has become more prevalent, possibly related to factors such as tuberculosis, sepsis, and tenofovir use, whereas there has been a decrease in the prevalence of classic HIVAN. Diana and colleagues¹³⁵ also reported an increase in the prevalence of FSGS (not otherwise specified). Kidney outcomes were poorest in patients with HIVAN, possibly related to late presentation.

The widespread use of ART has led to substantial reductions in the incidence of HIVAN and HIVAN-related kidney failure,¹³⁶ but there have been conflicting reports on the benefit of ART in patients with HIV-ICD. Szczech and colleagues¹³⁷ reported that lesions other than HIVAN did not benefit from ART. Several African studies have demonstrated improvements in kidney function in patients who have initiated ART for HIV-related kidney disease. These include the DART study in Uganda and Zimbabwe,¹³⁸ the study by Peters and colleagues in Uganda,¹³⁹ the study by Mpondo and colleagues in Tanzania,¹⁴⁰ and the South African studies by Wearne and colleagues¹³³ in Cape Town and Fabian and colleagues¹⁴¹ in Johannesburg. The South African studies included kidney biopsy data and, unlike the study by Szczech and colleagues,¹³⁷ demonstrated benefits in both HIVAN and HIV-ICD. In contrast, Inusah and colleagues¹⁴² found that HIV infection had no impact on kidney outcomes in patients with immune complex MCGN. The authors suggested that ART was unlikely to influence the kidney outcomes in the light of their patients' late presentation and severe histopathological findings.

KIDNEY STONES

North African countries lie within a geographic Afro-Asian stone-forming belt where kidney stone disease is particularly common. In this region, all age groups are affected, with a male predominance. The high prevalence of urolithiasis may be related to high rates of consanguinity, hot climates, and dietary factors such as high intake of animal protein.¹⁴³

Conversely, sub-Saharan Africa has the lowest incidence of kidney stones compared with other regions worldwide. Contributing factors include reduced genetic predisposition, a younger population living in rural areas with lower ambient temperatures, less sedentary lifestyles, and lower rates of obesity. Moreover, diets tend to be rich in micronutrients with less protein and sugar and there is less use of lithogenic

medications like acetazolamide, lithium, warfarin, calcium, and vitamin D supplements.¹⁴⁴⁻¹⁴⁷

Kaestner and colleagues¹⁴⁸ examined the prevalence of metabolic abnormalities in 175 patients attending a kidney stone clinic in Cape Town, South Africa. The main abnormalities were hypocitraturia (61%), hypomagnesuria (41%), mild hypercalciuria (22%), and hyperuricosuria (20%). In 432 Moroccan children, Meiouet and colleagues¹⁴⁹ found that calcium oxalate stones were the most prevalent (52%) followed by struvite (18%), ammonium urate (10%), and carbapatite stones (9%). Calcium oxalate accounted for 72% of stones in girls and 43% in boys; struvite stones were more frequent in boys than in girls (22% vs. 9%), as were calcium phosphate stones (12% vs. 4.6%).

There is a paucity of African data on CKD or kidney failure resulting from kidney stone disease. In Tunisia, 2.7% of patients on chronic dialysis have stones as the etiology.¹⁵⁰ In Sudan, 10% to 12% of patients on dialysis have obstructive uropathy as the etiology of their kidney failure.^{151,152} In Libya, the prevalence was 5%¹⁵³ and in Egypt it was 4% for stones and 2% for obstructive uropathy.⁹⁶ In South Africa, only 1.7% of patients on KRT had obstruction and/or reflux as the primary kidney disease.⁹⁴ There were few patients for whom there was specific mention of calculi, uric acid nephropathy, cystinosis, hyperoxaluria, or hypercalcemia.

SICKLE CELL DISEASE

Africa carries approximately 75% of the global burden of sickle cell disease, mainly due to a high prevalence of sickle cell trait in West Africa.¹⁵⁴ It is estimated that 25% of Nigerian adults are carriers. The presence of sickle cell trait protects against the severe complications of *Plasmodium falciparum* malaria, which is endemic in the region.

Kidney involvement is common and more severe in homozygous disease (sickle cell anemia, HbSS) than in the compound heterozygous forms; however, even sickle cell trait is associated with increased risk of CKD.¹⁵⁵ The renal involvement contributes to a diminished life expectancy, accounting for 16% to 18% of the mortality. Even patients who are able to access KRT have a mortality rate several-fold higher than patients without sickle cell disease.¹⁵⁶

Several gene variants have been implicated in the development of sickle cell nephropathy including *HBA1*, *HBA2*, *APOL1*, *HMOX1*, and *ACKR1*.¹⁵⁷ Hemolysis, vaso-occlusion, and ischemia-reperfusion injury are the mechanisms underlying the clinical manifestations of sickle cell disease. The pathophysiology of the renal involvement involves the coexistence of cortical hyperperfusion, medullary hypoperfusion, and an increased vasoconstrictive response to systemic and regional stress.¹⁵⁶ Red blood cells sickle in the renal medulla because of its hypoxic, acidotic, and hyperosmolar conditions.

Clinical manifestations range from hematuria and loss of concentrating ability to acute kidney injury, CKD, and renal medullary carcinoma. The most common glomerular lesion is FSGS. Hyperfiltration and impaired concentrating ability may occur in infancy, albuminuria may develop in childhood and progress thereafter, and in older adults the risk of CKD and kidney failure increases. Hematuria and AKI can occur at any age.¹⁵⁶ Hypertension was previously considered uncommon in patients with sickle cell disease,¹⁵⁸ but more recent evidence suggests that its prevalence may be higher, with reported rates reaching 19%.¹⁵⁹

A Nigerian case series by Arogundade and colleagues¹⁶⁰ reported that 37% of their patients with sickle cell disease had CKD as defined by persistent proteinuria, hematuria, and/or reduction in GFR. Another Nigerian study, among 200 stable patients, reported that 54% had albuminuria and only 6% had both albuminuria and an eGFR <60 mL/min per 1.73 m².¹⁶¹ In the Democratic Republic of Congo, a study of 98 stable patients found that 87% had albuminuria and only 4% had a low GFR.¹⁶²

Established therapies include hydroxyurea and red blood cell transfusions, as well as inhibitors of the renin-angiotensin-aldosterone system (RAS). A systematic review published in 2021¹⁶³ identified only one randomized controlled trial and concluded that there was not enough evidence that RAS inhibitors reduce proteinuria in people with sickle cell disease. Due to the risk of sickling crisis caused by potential hypovolemia, diuretics are typically avoided as a treatment option. When managing anemia, erythropoiesis-stimulating agents can safely be used in conjunction with hydroxyurea.¹⁶⁴ Recently, several newer therapies have been approved for use in sickle cell disease including L-glutamine, pyruvate kinase inhibitors, and gene therapies using modified autologous stem cells.¹⁶⁵

Data on African patients with sickle cell disease receiving KRT are limited. The Ghanaian Renal Registry reported that only 0.3% of treated patients had sickle cell disease. Dialysis does not appear to increase the risk of vasoocclusive crisis,¹⁶⁶ but there is an increased risk of vascular access failure,¹⁶⁷ possibly related to coagulopathy and endothelial dysfunction.¹⁵⁷ When performing hemodialysis, avoiding excessive ultrafiltration, using low dialysate temperatures, and using high dialysate sodium concentrations may reduce the risk of vaso-occlusive crises.¹⁵⁷ Peritoneal dialysis may be the preferred treatment modality as it avoids the need for vascular access for dialysis and reduces the risk of excessive volume depletion.¹⁵⁷ Kwarteng-Siaw and colleagues¹⁶⁸ have reported a survival advantage for patients treated with peritoneal dialysis. Kidney transplantation for patients with sickle cell disease remains a viable option, with survival rates comparable with those of patients with diabetes.¹⁶⁹

GENETIC DISORDERS

Participation of Africans in genomics research has been limited, with reasons for this including a shortage of African scientists with genomic research expertise, lack of research infrastructure, limited computational expertise and resources, and inadequate support by governments.¹⁷⁰ Adedokun and colleagues¹⁷¹ analyzed genetics publications from sub-Saharan Africa for the period 2004–2013 and found that most of the publications involved populations from South Africa (31.1%), Ghana (10.6%), and Kenya (7.5%). Less than half of the publications had a first author from an African institution. Ilori and colleagues¹⁷² have reviewed genetics research in low-resource settings and reported that only 5% of genome-wide association studies (GWAS) have been conducted in low-resource settings, which are home to nearly 60% of the world's population. Africans represent only 8.7% of all publicly available next-generation sequencing data, despite accounting for 15.4% of the world's population.¹⁷²

The Human Heredity and Health in Africa (H3Africa, <https://h3africa.org/>) initiative is a large-scale collaboration

between the U.S. National Institutes of Health, the U.K.-based Wellcome Trust, and the African Society of Human Genetics, which has begun to address some of these issues by focusing strongly on capacity building, as well as specific scientific goals.¹⁷⁰ The project is training African clinical research personnel and genomic investigators, establishing genomic research laboratories, and conducting international-level genetic and translational research. Three biorepositories have been established in Nigeria, Uganda, and South Africa for DNA and other biosamples. The H3Africa Kidney Disease Research Network (H3AKDRN) studies the genetic and environmental factors associated with CKD including the role of the African “risk” variants of the *APOL1* gene and genetic factors involved in sickle cell disease. H3AKDRN has recruited more than 10,000 patients and healthy controls in Ghana, Nigeria, Ethiopia, and Kenya.^{173,174} Of these, 3000 have had a GWAS using the new H3Africa Illumina chip and 750 have had kidney biopsies.¹⁷⁵ So far, they have discovered that participants carrying two *APOL1* risk alleles have a 45% higher risk of developing CKD compared with those with one or no risk alleles. Nearly one-quarter of healthy controls from Ghana and Nigeria carry the two risk alleles, and one-third of participants with CKD carry these same alleles.

The Role of Consanguineous Marriage

A consanguineous marriage is defined as a union between two individuals who are related as second cousins or closer. This practice is common throughout Arab countries and contributes to the increase in genetic disorders, especially autosomal recessive disorders, seen in North Africa (see Chapter 76). For example, the high rates of kidney stone disease seen in the region are at least partly due to consanguinity, which results in higher rates of predisposing conditions such as primary hyperoxaluria and cystinosis.¹⁴³ Jaouad and colleagues¹⁷⁶ studied the rates of consanguineous marriages in 176 Moroccan families with autosomal recessive diseases. While the overall prevalence of consanguinity in Morocco was reported to be 15%, the rate among the families with autosomal recessive disorders was 59%.

Cystic Kidney Disease

Worldwide, autosomal dominant polycystic kidney disease (ADPKD) is a common cause of CKD and kidney failure, but the data from African populations are sparse. Bourqia¹⁷⁷ reported data on ADPKD in 308 Moroccan families. The mean age at diagnosis was 46 years, and pain was the most frequent presenting symptom. Loss of kidney function was present in 17% and hepatic cysts in 18%. In Benin, Laleye and colleagues¹⁷⁸ described 70 patients with a mean age of 47 years. A positive family history was obtained in 47%, and flank pain was the most common presenting symptom. Mutations in the polycystin 1 gene (*PKD1*) were identified including several variants not previously described. In Senegal, a cohort of 53 patients has been described,¹⁷⁹ as has a case where a new *PKD1* mutation was identified.¹⁸⁰ A Ghanaian study¹⁸¹ of 82 patients reported flank pain (31%) and leg swelling (18%) as the most common symptoms. CKD was present in 82% of the patients, with 20% having stage 5 CKD. In Nigeria, 41 patients with ADPKD were seen over a 15-year period.¹⁸² Nocturia (78%) and flank pain (68%) were common symptoms, and a family history of ADPKD was present in 56%. Liver cysts (32%) and aortic regurgitation (22%) were the predominant extrarenal

manifestations. The median eGFR was 34 mL/min per 1.73 m² at presentation, and many progressed to kidney failure, with 63% starting dialysis. In the Seychelles, ADPKD has been seen almost exclusively in people of Caucasian extraction, even though they comprise only 30% of the population. This may be the consequence of a founder effect.¹⁸³

Autosomal recessive polycystic kidney disease (ARPKD) is caused by mutations in the *PKHD1* gene and has a worldwide incidence of 1:20,000, corresponding to a heterozygote frequency of 1:70. Among the Afrikaner population of South Africa, the live-birth rate and carrier rate have been estimated at 1:11,000 and 1:53, respectively.¹⁸⁴ Lambie and colleagues¹⁸⁵ studied a cohort of patients from 36 Afrikaner families, finding that 27 patients, from 24 families, were homozygous for the *p.M627K* substitution, providing strong evidence of a founder mutation.

Nephronophthisis (NPHP) is an autosomal recessive cystic kidney disease that is a leading genetic cause of kidney failure in children. In contrast to ADPKD, the kidneys are not enlarged, cysts are present mostly at the corticomedullary junction, and there is prominent tubulointerstitial fibrosis.¹⁸⁶ Soliman and colleagues¹⁸⁷ studied 20 Egyptian children with NPHP, reporting a mean age at diagnosis of 88 months and a mean age at the onset of symptoms of 44 months. Patients were categorized as 75% juvenile NPHP, 5% infantile NPHP, and 20% Joubert syndrome-related disorders.

Metabolic Diseases

Primary hyperoxaluria type 1 (PH1) is an autosomal recessive metabolic disease where there is increased endogenous production of oxalate (see [Chapters 40 and 71](#)). Progressive kidney failure results from excessive urinary oxalate excretion and renal deposition of calcium oxalate. Disease-causing mutations of the alanine glyoxylate aminotransferase (*AGXT*) gene have been identified in patients from Tunisia¹⁸⁸ and Egypt.¹⁸⁹

Cystinosis is an autosomal recessive lysosomal storage disease caused by mutations in the *CTNS* gene. A study of 20 South African children with nephropathic infantile type cystinosis, characterized by proximal tubule dysfunction,¹⁹⁰ found that 19 children had a G>A mutation in intron 11 and that 16 were homozygous for this mutation.

Hereditary tyrosinemia type 1 (HT1) is caused by a mutation that results in a deficiency of fumarylacetoacetate hydrolase, leading to tyrosine accumulation. Kidney-related manifestations include nephromegaly, nephrocalcinosis, Fanconi syndrome, kidney stones, and CKD. El-Karakasy and colleagues¹⁹¹ studied 22 affected Egyptian children and found that 64% had rickets accompanied by hypophosphatemia, while 77% had nephromegaly.

Familial Mediterranean fever is an autosomal recessive disease that primarily affects populations surrounding the Mediterranean and is characterized by recurrent attacks of fever and serosal inflammation. Amyloidosis is a serious complication of the disease and may result in renal failure. The disease is caused by mutations in the familial Mediterranean fever gene (*MEFV*), which encodes pyrin, a protein involved in regulating inflammatory responses. Mutations may cause gain of function so that pyrin can initiate inflammation in the absence of a toxin or infection. *MEFV* mutations have been identified in Tunisian patients.¹⁹² Amyloidosis is more common in North African

patients who are homozygous for the *M694V* mutation, and the risk is also increased in male patients and those bearing the polymorphism a/a in the *SAAI* gene.¹⁹³

Tubular Disorders

Familial hypomagnesemia with hypercalciuria and nephrocalcinosis caused by a novel claudin16 mutation has been reported in an Egyptian family.¹⁹⁴ In South Africa, new mutations of the *SLC12A3* gene, which encodes the sodium-chloride cotransporter on the apical membrane of the distal convoluted tubule, have been reported in a family with Gitelman syndrome who presented with hypokalemia and unusual cravings for salt and vinegar.¹⁹⁵ Mutations of the epithelial sodium channel (ENaC) have also been identified in South African patients in association with hypertension^{196,197} and preeclampsia.¹⁹⁸ In Tunisia, mutations of the H⁺-ATPase pump have been identified in children with primary distal renal tubular acidosis, often associated with sensorineural deafness.¹⁹⁹

Glomerular Diseases

Mutations of the nephrin gene (*NPHS1*) have been discovered in North African patients with congenital nephrotic syndrome of the Finnish type.²⁰⁰ A Moroccan infant with steroid-resistant nephrotic syndrome had a mutation in the *ARHGDI* gene, which encodes Rho-GDP dissociation inhibitor 1. This protein modulates signaling through Rho GTPases and thereby regulates cell motility.²⁰¹

Nandlal and colleagues²⁰² examined causal genetic mutations associated with steroid-resistant nephrotic syndrome in 70 South African children. Potentially pathogenic mutations were identified in 39% of the children, with 21% carrying the *NPHS2 p.V260E* homozygous mutation, which is linked to biopsy-proven FSGS. Additionally, 17% of the children had heterozygous mutations involving the *INF2*, *CD2AP*, and *TRPC6* genes, which have an autosomal dominant inheritance mode.

CKD OF UNCERTAIN ETIOLOGY (SEE ALSO "CLASSIFICATION AND MANAGEMENT OF CHRONIC KIDNEY DISEASE")

CKD of uncertain etiology (CKDu) describes CKD not attributable to traditional risk factors such as diabetes, hypertension, or HIV infection. CKDu "hotspots" have been reported from several African countries. For example, in Tunisia, chronic interstitial nephropathy has been linked to the contamination of food with ochratoxin,²⁰³ while in Tanzania, urban residence was a strong risk factor for the presence of CKD.²⁰⁴ In the El-Minia region of Egypt, kidney failure with an unknown etiology in rural farm workers has been associated with environmental factors, such as drinking unsafe water, exposure to pesticides, and using herbal medications.²⁰⁵ In Malawi, however, a study of CKDu in rural and urban adults without diabetes, hypertension, or proteinuria found a low prevalence of 0.2% (CKDu is discussed in more detail in [Chapters 74 and 83](#)).²⁰⁶

KIDNEY FAILURE AND KIDNEY REPLACEMENT THERAPY

Wide variations in the reported prevalence of treated kidney failure seen in developing countries are mainly a reflection

of differing access to dialysis and transplantation rather than differences in the burden of disease. According to the latest ISN-GKHA,¹⁶ hemodialysis was available in all of the 41 responding African countries, but only half had the capacity to provide regular, thrice-weekly treatments. The capacity to provide adequate chronic peritoneal dialysis (3–4 exchanges daily) was available in 8 countries, and kidney transplantation services were available in 13 countries.

In most African countries, patients carry the cost of KRT and few can afford dialysis beyond the first few months. Ashuntantang and colleagues²⁰⁷ conducted a systematic review of the outcomes in sub-Saharan patients with kidney failure who required maintenance dialysis. They reported a pooled treatment discontinuation rate in incident adult patients of 84%, with only 10% still on dialysis at 3 months. Only 61% of children with kidney failure received one or more dialysis sessions, and only 35% remained on dialysis for at least 3 months. This attrition usually occurred within the first 2 weeks of starting dialysis and was mainly the result of an inability to pay for the treatment.

The serious moral distress experienced by nephrologists working in low-resource settings in sub-Saharan Africa has been highlighted by Ashuntantang and colleagues.²⁰⁸ The inability of patients to pay for dialysis and the lack of resources and infrastructure in many hospitals force nephrologists to practice rationing and sometimes to choose between patients, usually without the benefit of clear institutional guidelines. Most nephrologists felt burdened by ethical dilemmas, half of them reported regretting their choice of profession, and one-quarter had considered emigration.

According to the ISN-GKHA,¹⁶ conservative/supportive kidney care established through shared decision making, choice-restricted supportive care where resource constraints limit access, and choice-restricted supportive care where no resource constraints limit access was available in 15 (38%), 13 (33%), and 10 (25%) of the 41 responding African countries, respectively. South African health care professionals have compiled a practical consensus guideline to help care providers improve the supportive management of their patients with kidney failure, whether they are able to access KRT or not.²⁰⁹ The latest ISN-GKHA¹⁶ data for the Africa region indicated a median annual cost for in-center hemodialysis of US\$13,793, for continuous ambulatory peritoneal dialysis of US\$14,192, and for the first year of kidney transplantation of US\$20,714.

The launch of the African Renal Registry in 2015²¹⁰ was an important initiative of the African Association of Nephrology, which provided access to a data capture platform for African countries wishing to start a renal registry. It uses the web-based platform of the South African Renal Registry. Botswana, Burundi, Ghana, Kenya, Nigeria, and Zambia have formally joined the African Renal Registry. Another important initiative is the ISN's SHARing Expertise to support the set-up of Renal Registries (SharE-RR) project.²¹¹ SharE-RR develops resources to help establish or develop renal registries in emerging countries. The latest ISN-GKHA survey¹⁶ indicated that 12 of the 41 responding African countries had a dialysis registry (Angola, Botswana, Côte d'Ivoire, Egypt, Ghana, Kenya, Malawi, Mauritius, Morocco, Niger, South Africa, and Zambia) and 5 had a kidney transplant registry (Botswana, Egypt, Kenya, Mauritius, and South Africa).

Table 75.4 summarizes African publications that have recently reported data on KRT. Formal registry reports are available for South Africa,⁹⁴ Egypt,⁹⁶ and Ghana.⁹⁵ The South African Renal Registry is the only one that publishes country-wide data and has just published its 10th consecutive annual report.⁹⁴ In 2021, there was an overall prevalence of 147 pmp; in the public health care sector, which serves 85% of the South African population, the prevalence was 45 pmp, as compared with 736 pmp in the private sector. Disparities between different provinces were highlighted, as were disparities related to ethnicity, with Blacks being the most underrepresented group.

Recent data on the survival of African patients starting KRT have been published from South Africa, Zambia, and Cameroon. In studies using South African Renal Registry data, Jardine and colleagues²¹² reported good overall 1-year survival of 90% and Thapa and colleagues²¹³ reported 1-year survival of 86% in elderly patients. Muller and colleagues²¹⁴ have reported good outcomes for HIV-positive-to-positive transplantation, comparable with those of HIV-negative recipients, with a 1-year patient survival rate of 84% and 5-year survival of 74%. The rates for graft survival (censored for death with a functioning graft) were 93% and 84%, respectively.

In Zambian patients starting chronic dialysis, Banda and colleagues²¹⁵ recorded a mortality rate at 90 days of 12%. The main causes of this early mortality were cardiovascular disease (42%) and catheter-related infections (32%). Teuwafeu and colleagues²¹⁶ studied a cohort of 81 elderly patients starting dialysis in Cameroon and reported a 1-year survival rate of 41%. The most common causes of death were sudden death (42%), infection/sepsis (21%), and dialysis withdrawal (17%).

Several African studies have examined the quality of life (QOL) of patients on KRT. These have mainly used standardized questionnaires such as the Kidney Disease Quality of Life–Short Form (KDQOL-SF) and the KDQOL-36 questionnaire. In South Africa, Okaka and colleagues²¹⁷ found that patients on peritoneal dialysis in Johannesburg had lower QOL scores than healthy controls, with better scores in younger patients, those with a duration on PD of less than 4 years, and those with higher incomes. In Cape Town, Tannor and colleagues²¹⁸ used a mixed-methods approach and found that patients on peritoneal dialysis (compared with those on hemodialysis) experienced greater symptom burden and greater limitations related to their dialysis modality, especially with regards to social functioning. The fear of developing peritonitis caused many patients to sacrifice social interactions to be able to do their exchanges at home and at specified times. In Malawi, Masina and colleagues²¹⁹ found the QOL scores of their hemodialysis patients to be low, particularly in the case of physical health scores. Similar findings were reported from Kenya²²⁰ and in Tanzania, where receiving fewer than three hemodialysis sessions per week and having a urea reduction ratio of <65% were associated with lower scores.²²¹ Bagasha and colleagues²²² undertook a cultural adaptation and validation of the KDQOL-SF questionnaire in Uganda and subsequently found low QOL scores in 94% of their patients with stage 5 CKD including those receiving hemodialysis and those not treated with dialysis.²²³ There was no difference in QOL scores between these two groups. In Morocco, El Filali and colleagues²²⁴ reported high rates

Table 75.4 Recent Publications Reporting Data on Kidney Replacement Therapy from African Countries^a

Author and Year	Country	Last Data	Key Data Reported
Sumaili et al., 2021 ²³⁸	Democratic Republic of the Congo	2019	KRT prevalence 3 pmp. KRT modality: HD 98%, PD 2%, TX 0%. Vascular access: AVF 9%, TnC 61%, TmpC 30%. Other: HCV+ 17%, HBV+ 0%, HIV+ 0%.
Hassaballa et al., 2020 ⁹⁶	Egypt	2020	Cause: hypertension 41%, diabetes 14%, unknown 13%. Dialysis modality: HD 100%. Vascular access: AVF 87%, AVG 5%, TnC 4%, TmpC 4%. Other: HCV+ 31%, HBV+ 1.6%, HIV+ 0.2%.
Mengistu et al., 2022 ²³⁹	Ethiopia	2021	Dialysis prevalence 10 pmp. Dialysis modality: HD 100%. Vascular access: AVF 72%, AVG 3%, TnC 10%, TmpC 16%. Cause: hypertension 38%, unknown 29%, diabetes 9%.
Boima et al., 2021, ⁹⁵ Okyere et al., 2021 ²⁴⁰	Ghana	2017	KRT modality: HD 96%, TX 4%, only 2 patients on PD. Vascular access: AVF 27%, TnC 19%, TmpC 54%. Other: HCV+ 0.8%, HBV+ 3.9%, HIV+ 3.8%.
Abdulla et al., 2023 ²⁴¹	Maldives	2022	Prevalence 704 pmp. Cause: hypertension 47%, diabetes 24%, glomerulonephritis 20%. Dialysis modality: HD 100%. Vascular access: AVF 86%, TnC 14%.
Ip Min Wan et al., 2021 ²⁴²	Mauritius	2020	KRT prevalence 1386 pmp. KRT modality: HD 90%, TX 10%, only 1 patient on PD. Vascular access: AVF 87%, TnC 2%, TmpC 11%. Other: HCV+ 0.8%, HBV+ 0.2%, HIV+ 0.5%.
Okoye et al., 2022 ²⁴³	Nigeria	2022	Dialysis prevalence 17 pmp. Dialysis modality: HD 100%. Vascular access: CVC 90% (mostly TmpC), AVF 9%, AVG 1%.
Niang et al., 2020, ⁹⁸ Faye et al., 2023 ²⁴⁴	Senegal	2021	Dialysis prevalence 67 pmp. Cause: unknown 37%, hypertension 33%, glomerulonephritis 11%. Dialysis modality: HD 96%, PD 4%. Vascular access: AVF 65%, AVG 2%, TnC 28%, TmpC 4%.
Davids et al., 2023 ⁹⁴	South Africa	2021	KRT incidence 14 pmp, KRT prevalence 147 pmp. Cause: hypertension 38%, unknown 30%, diabetes 13%. KRT modality: HD 71%, PD 10%, TX 19%. Other: HCV+ 2.2%, HBV+ 2.3%, HIV+ 12%.
Kalyesubula et al., 2022 ²⁴⁵	Uganda	2020	KRT prevalence 52 pmp. Dialysis modality: HD 100%.

^aClinical data relate to prevalent patients.

AVF, Arteriovenous fistula; AVG, arteriovenous graft; CVC, central venous catheter; HBV+, seropositive for hepatitis B virus; HCV+, seropositive for hepatitis C virus; HD, hemodialysis; HIV+, seropositive for human immunodeficiency virus; KRT, kidney replacement therapy; PD, peritoneal dialysis; pmp, per million population; TX, transplant; TmpC, temporary catheter; TnC, tunneled catheter.

of major depressive and anxiety disorders (34% and 25%, respectively) in patients receiving chronic hemodialysis. Major depressive episodes were associated with living alone, the presence of pain, and anxiety disorders. Boudida and colleagues²²⁵ developed a Moroccan version of the KDQOL-SF and demonstrated its validity and reliability in Moroccan patients. In Egypt, Abdelfatah and colleagues²²⁶ found that younger patients had better QOL scores and that there was no difference in overall QOL with different modalities of KRT.

CONCLUSION

Africa faces many health challenges as its rapidly growing populations struggle with the burdens of infections,

noncommunicable disease, pregnancy-related disease, and injuries. These factors are driving an epidemic of kidney disease, which often affects young, economically active individuals and has serious economic consequences for their families and communities. In most African countries, financial and human resources are inadequate to meet this challenge and many African governments have not yet recognized CKD as a national health priority.

However, several successful initiatives provide hope for the future. The ISN and International Society for Peritoneal Dialysis Fellowship programs continue to make a big contribution to the training of nephrology health care workers in Africa, with most of these fellowships now occurring at centers within the Africa region. Events such as World Kidney Day are creating awareness of kidney disease among the public and health professionals, and the outcomes

for patients with AKI are improving as a result of initiatives such as the Saving Young Lives project. Finally, research projects such as the H3Africa project and the African Renal Registry have the potential to generate important insights into kidney disease on the continent, improve the outcomes of African patients with kidney disease, and provide valuable training platforms for African researchers.

 Visit Elsevier eBooks+ (eBooks.Health.Elsevier.com) for a complete set of references.

KEY REFERENCES

3. Africa overview. 2022 World Population Data Sheet: Population Reference Bureau; 2022.
12. HIV Statistics, Globally and by WHO Region, 2023. World Health Organisation; 2023, at <https://cdn.who.int/media/docs/default-source/hq-hiv-hepatitis-and-stis-library/j0294-who-hiv-epi-factsheet-v7.pdf>. Accessed September 29, 2023.
13. Bamgboye EL, Omiye JA, Afolaranmi OJ, et al. COVID-19 pandemic: is Africa different? *J Natl Med Assoc*. 2020.
14. Chothia MY, Barday Z, Nel J, Davids MR. Impact of COVID-19 on access to chronic kidney replacement therapy in the public sector of Western Cape Province, South Africa. *S Afr Med J*. 2021;111:13442.
16. Bello AK, Okpechi IG, Levin A, et al. *ISN–Global Kidney Health Atlas: A Report by the International Society of Nephrology: An Assessment of Global Kidney Health Care Status Focusing on Capacity, Availability, Accessibility, Affordability and Outcomes of Kidney Disease*. Brussels, Belgium: International Society of Nephrology; 2023.
21. Okpechi IG, Eddy AA, Jha V, Jacob T, Dupuis S, Harris DC. Impact of training nephrologists from developing nations and strategies for sustaining a training program in its fourth decade. *Kidney Int*. 2021;99:1073–1076.
22. Mehta RL, Burdmann EA, Cerda J, et al. Recognition and management of acute kidney injury in the International Society of Nephrology 0by25 Global Snapshot: a multinational cross-sectional study. *Lancet*. 2016;387:2017–2025.
24. Batte A, Shahrin L, Claire-Del Granado R, Luyckx VA, Conroy AL. Infections and acute kidney injury: a global perspective. *Semin Nephrol*. 2023;151466.
25. Olowu WA, Niang A, Osafo C, et al. Outcomes of acute kidney injury in children and adults in sub-Saharan Africa: a systematic review. *Lancet Glob Health*. 2016;4:e242–e250.
27. *World Malaria Report 2022*. Geneva, Switzerland: World Health Organization; 2022.
32. Davidson B, Bajpai D, Shah S, et al. Pregnancy-associated acute kidney injury in low-resource settings: progress over the last decade. *Semin Nephrol*. 2022;42:151317.
33. Ekrikpo UE, Adejumo OA, Akpan EE, et al. The prevalence of acute kidney injury in women with hypertensive disorders of pregnancy in Africa: a systematic review and meta-analysis. *Afr J Nephrol*. 2023;26:95–105.
34. Shalaby AS, Shemies RS. Pregnancy-related acute kidney injury in the African continent: Where do we stand? A systematic review. *J Nephrol*. 2022;35:2175–2189.
36. Stanifer JW, Kilonzo K, Wang D, et al. Traditional medicines and kidney disease in low- and middle-income countries: opportunities and challenges. *Semin Nephrol*. 2017;37:245–259.
49. Smoyer WE, Finkelstein FO, McCulloch MI, Carter M, Brusselmanns A, Feehally J. “Saving Young Lives” with acute kidney injury: the challenge of acute dialysis in low-resource settings. *Kidney Int*. 2016;89:254–256.
51. McCulloch MI, Nourse P, Argent AC. Use of locally prepared peritoneal dialysis (PD) fluid for acute PD in children and infants in Africa. *Perit Dial Int*. 2020;40:441–445.
60. Kalyesubula R, Conroy AL, Calice-Silva V, et al. Screening for kidney disease in low- and middle-income countries. *Semin Nephrol*. 2022;151315. Elsevier.
65. Holness JL, van der Westhuizen DJ, Davids MR, Warwick JM. Measurement of glomerular filtration rate and its current status in African countries. *Afr J Nephrol*. 2022;25:74–81.
76. Fabian J, George JA, Etheredge HR, et al. Methods and reporting of kidney function: a systematic review of studies from sub-Saharan Africa. *Clin Kidney J*. 2019;12:778–787.
81. Stanifer JW, Jing B, Tolan S, et al. The epidemiology of chronic kidney disease in sub-Saharan Africa: a systematic review and meta-analysis. *Lancet Global Health*. 2014;2:e174–e181.
86. Kaze AD, Ilori T, Jaar BG, Echouffo-Tcheugui JB. Burden of chronic kidney disease on the African continent: a systematic review and meta-analysis. *BMC Nephrol*. 2018;19:125.
87. Abd EHS, Bolognani D, D’Arrigo G, Dounousi E, Tripepi G, Zoccali C. Prevalence and burden of chronic kidney disease among the general population and high-risk groups in Africa: a systematic review. *BMJ Open*. 2018;8:e015069.
88. Kayange NM, Smart LR, Tallman JE, et al. Kidney disease among children in sub-Saharan Africa: systematic review. *Pediatr Res*. 2014;77:272–281.
90. *IDF Diabetes Atlas 10th edition*. Brussels, Belgium: International Diabetes Federation; 2021.
92. Wagnew F, Eshetie S, Kibret GD, et al. Diabetic nephropathy and hypertension in diabetes patients of sub-Saharan countries: a systematic review and meta-analysis. *BMC Res Notes*. 2018;11:1–7.
93. Ajayi SO, Ekrikpo UE, Ekanem AM, et al. Prevalence of chronic kidney disease as a marker of hypertension target organ damage in Africa: a systematic review and meta-analysis. *Int J Hypertens*. 2021;2021.
94. Davids MR, Marais N, Sebastian S, Jardine T, Jacobs JC. South African renal registry annual report 2021. *Afr J Nephrol*. 2023;26:83–94.
95. Boima V, Tannor EK, Osafo C, et al. The Ghana Renal Registry—a first annual report. *Afr J Nephrol*. 2021;24:19–24.
96. Hassaballa M, El-Wakil H, Elsharkawy M, et al. Egyptian renal data system (ERDS) 2020: an annual report of end-stage kidney disease patients on regular hemodialysis. *ESNT*. 2022;22:1–28.
103. Esmail AM, Bates WD, Amirali MH, Jardine T, Davids MR. Patterns of biopsy-proven kidney disease amongst South African adults from 1995 to 2017. *Afr J Nephrol*. 2023;26:9–16.
104. Vermeulen A, Menezes C, Mashabane M, et al. Patterns of renal disease: a 30-year renal biopsy study at Chris Hani Baragwanath academic hospital, Soweto, Johannesburg, South Africa. *S Afr Med J*. 2019;109:486–492.
111. Okpechi IG, Ameh OI, Bello AK, Ronco P, Swanepoel CR, Kengne AP. Epidemiology of histologically proven glomerulonephritis in Africa: a systematic review and meta-analysis. *PLoS One*. 2016;11:e0152203.
113. Uwaezuoke SN, Ndu IK, Mbanefo NR. Prevalence rates of histopathologic subtypes associated with steroid resistance in childhood nephrotic syndrome in Sub-Saharan Africa: a systematic review. *Int J Nephrol Renovasc Dis*. 2019;167–176.
127. Kasembeli AN, Duarte R, Ramsay M, et al. APOL1 risk variants are strongly associated with HIV-associated nephropathy in black South Africans. *J Am Soc Nephrol*. 2015.
130. Koech M, Owiti M, Owino-Ong’or W, et al. Absence of HIV-associated nephropathy among antiretroviral naïve adults with persistent albuminuria in Western Kenya. *Kidney Int Rep*. 2017;2:159–164.
134. Wearne N, Manning K, Price B, et al. The evolving spectrum of kidney histology in HIV-positive patients in South Africa. *Kidney Int Rep*. 2023;8:1087–1096.
135. Diana NE, Davies M, Mosiane P, Vermeulen A, Naicker S. Clinicopathological correlation of kidney disease in HIV infection pre- and post-ART rollout. *PLoS One*. 2022;17:e0269260.
141. Fabian J, Naicker S, Goetsch S, Venter WD. The clinical and histological response of HIV-associated kidney disease to antiretroviral therapy in South Africans. *Nephrol Dial Transplant*. 2013;28:1543–1554.
154. Arogundade FA, Hassan MO, Omotoso BA, et al. Spectrum of kidney diseases in Africa: malaria, schistosomiasis, sickle cell disease, and toxins. *Clin Nephrol*. 2016;86:S53–S60.
157. Ataga KI, Saraf SL, Derebail VK. The nephropathy of sickle cell trait and sickle cell disease. *Nat Rev Nephrol*. 2022;18:361–377.
171. Adedokun BO, Olopade CO, Olopade OI. Building local capacity for genomics research in Africa: recommendations from analysis of publications in Sub-Saharan Africa from 2004 to 2013. *Glob Health Action*. 2016;9:31026.
175. Adu D, Ojo A, H3A-KDRN Investigators. Overview of the human Heredity and Health in Africa Kidney Disease Research Network (H3A-KDRN). *Kidney360*. 2021;2:129.

207. Ashuntantang G, Osafo C, Olowu WA, et al. Outcomes in adults and children with end-stage kidney disease requiring dialysis in sub-Saharan Africa: a systematic review. *Lancet Glob Health*. 2017;5:e408–e417.
208. Ashuntantang G, Miljeteig I, Luyckx VA. Bedside rationing and moral distress in nephrologists in sub-Saharan Africa. *BMC Nephrol*. 2022;23:1–14.
210. Davids MR, Eastwood JB, Selwood NH, et al. A renal registry for Africa: first steps. *Clin Kidney J*. 2016;9:162–167.
212. Jardine T, Wong E, Steenkamp R, Caskey FJ, Davids MR. Survival of South African patients on renal replacement therapy. *Clin Kidney J*. 2020;13:782–790.
214. Muller E, Barday Z, Mendelson M, Kahn D. HIV-positive-to-HIV-positive kidney transplantation—results at 3 to 5 years. *N Engl J Med*. 2015;372:613–620.
218. Tannor EK, Archer E, Kapembwa K, Van Schalkwyk SC, Davids MR. Quality of life in patients on chronic dialysis in South Africa: a comparative mixed methods study. *BMC Nephrol*. 2017;18:4.
222. Bagasha P, Naitala R, Namukwaya E, Leng M, Katabira E, Namisango E. Cultural adaptation and validation of the Kidney Disease and Quality of Life-Short Form (KDQOL-SF™) version 1.3 questionnaire in Uganda. *Afr J Nephrol*. 2022;25:116–135.
224. El Filali A, Bentata Y, Ada N, Oneib B. Depression and anxiety disorders in chronic hemodialysis patients and their quality of life: A cross-sectional study about 106 cases in the northeast of morocco. *Saudi J Kidney Dis Transpl*. 2017;28:341–348.

REFERENCES

- 2022 World Population Data Sheet: Population Reference Bureau. 2022.
- World Population Prospects 2022: Summary of Results. United Nations, Department of Economic and Social Affairs; 2022. Population Division.
- Africa overview. 2022 World Population Data Sheet: Population Reference Bureau; 2022.
- Leveraging Resource Wealth during the Low Carbon Transition. *Africa's Pulse*, No. 27. Washington, DC: World Bank; 2023.
- Health Spending as Percent of GDP - Country Rankings [Data Source: The World Bank]. *TheGlobalEconomy.com*. 2020, at https://www.theglobaleconomy.com/rankings/health_spending_as_percent_of_gdp/Africa/. Accessed November 13, 2023.
- Deaths from Noncommunicable Diseases on the Rise in Africa. World Health Organization; 2022, at <https://www.afro.who.int/news/deaths-noncommunicable-diseases-rise-africa>. Accessed September 29, 2023.
- Noncommunicable Diseases Among Young People in Africa. Washington, DC: Population Reference Bureau; 2015.
- James PB, Bah AJ, Kabba JA, Kassim SA, Dalinjong PA. Prevalence and correlates of current tobacco use and non-user susceptibility to using tobacco products among school-going adolescents in 22 African countries: a secondary analysis of the 2013-2018 global youth tobacco surveys. *Arch Publ Health*. 2022;80:1–15.
- Marquez PV, Farrington JL. *The challenge of Non-communicable Diseases and Road Traffic Injuries in Sub-saharan Africa: An Overview*. Washington, DC: The World Bank; 2013.
- Road safety in the WHO African Region. *The Facts 2013*. Geneva, Switzerland: World Health Organization; 2013.
- World Health Organization. *World Health Statistics 2016: Monitoring Health for the SDGs*. World Health Organization; 2016.
- HIV Statistics, Globally and by WHO Region, 2023. World Health Organisation; 2023, at <https://cdn.who.int/media/docs/default-source/hq-hiv-hepatitis-and-stis-library/j0294-who-hiv-epi-factsheet-v7.pdf>. Accessed September 29, 2023.
- Bamgboye EL, Omiye JA, Afolaranmi OJ, et al. COVID-19 pandemic: is Africa different? *J Natl Med Assoc*. 2020.
- Chothia MY, Barday Z, Nel J, Davids MR. Impact of COVID-19 on access to chronic kidney replacement therapy in the public sector of Western Cape Province, South Africa. *S Afr Med J*. 2021;111:13442.
- Global health observatory (GHO) data. *Health Workforce*. World Health Organization; 2023, at <https://www.who.int/data/gho/data/themes/health-workforce>. Accessed October 2, 2023.
- Bello AK, Okpechi IG, Levin A, et al. *ISN–Global Kidney Health Atlas: A Report by the International Society of Nephrology: An Assessment of Global Kidney Health Care Status Focusing on Capacity, Availability, Accessibility, Affordability and Outcomes of Kidney Disease*. Brussels, Belgium: International Society of Nephrology; 2023.
- Jha V, Arici M, Collins AJ, et al. Understanding kidney care needs and implementation strategies in low- and middle-income countries: conclusions from a “Kidney Disease: Improving Global Outcomes” (KDIGO) Controversies Conference. *Kidney Int*. 2016;90:1164–1174.
- Eastwood JB, Conroy RE, Naicker S, West PA, Tutt RC, Plange-Rhule J. Loss of health professionals from sub-Saharan Africa: the pivotal role of the UK. *Lancet*. 2005;365:1893–1900.
- Harris DC, Dupuis S, Couser WG, Feehally J. Training nephrologists from developing countries: does it have a positive impact? *Kidney Int Suppl*. 2012;2:275–278.
- Ebeye T, Lee H. Down the brain drain: a rapid review exploring physician emigration from West Africa. *Glob Health Res Policy*. 2023;8:23.
- Okpechi IG, Eddy AA, Jha V, Jacob T, Dupuis S, Harris DC. Impact of training nephrologists from developing nations and strategies for sustaining a training program in its fourth decade. *Kidney Int*. 2021;99:1073–1076.
- Mehta RL, Burdmann EA, Cerda J, et al. Recognition and management of acute kidney injury in the International Society of Nephrology 0by25 Global Snapshot: a multinational cross-sectional study. *Lancet*. 2016;387:2017–2025.
- Mehta RL, Macedo E, Zhang J, ISN 0by25 Initiative Steering Committee. International society of Nephrology 0by25 global snapshot. Regional report for Africa 2015.
- Batte A, Shahrin L, Claire-Del Granado R, Luyckx VA, Conroy AL. Infections and acute kidney injury: a global perspective. *Semin Nephrol*. 2023;15:1466.
- Olowu WA, Niang A, Osafo C, et al. Outcomes of acute kidney injury in children and adults in sub-Saharan Africa: a systematic review. *Lancet Glob Health*. 2016;4:e242–e250.
- Bourial M. *Aspects épidémiologiques de l'insuffisance rénale aigüe au Maroc. Première enquête nationale*. Casablanca, Morocco: Université Hassan II de Casablanca; 2014.
- World Malaria Report 2022. Geneva, Switzerland: World Health Organization; 2022.
- Jha V, Parameswaran S. Community-acquired acute kidney injury in tropical countries. *Nat Rev Nephrol*. 2013;9:278–290.
- Mishra SK, Das BS. Malaria and acute kidney injury. *Semin Nephrol*. 2008;28:395–408.
- Wyatt CM, Arons RR, Klotman PE, Klotman ME. Acute renal failure in hospitalized patients with HIV: risk factors and impact on in-hospital mortality. *AIDS*. 2006;20:561–565.
- Boswell MT, Rossouw TM. Approach to acute kidney injury in HIV-infected patients in South Africa. *South Afr J HIV Med*. 2017;18:a714.
- Davidson B, Bajpai D, Shah S, et al. Pregnancy-associated acute kidney injury in low-resource settings: progress over the last decade. *Semin Nephrol*. 2022;42:1513–17.
- Ekrikpo UE, Adejumo OA, Akpan EE, et al. The prevalence of acute kidney injury in women with hypertensive disorders of pregnancy in Africa: a systematic review and meta-analysis. *Afr J Nephrol*. 2023;26:95–105.
- Shalaby AS, Shemies RS. Pregnancy-related acute kidney injury in the African continent: Where do we stand? A systematic review. *J Nephrol*. 2022;35:2175–2189.
- Luyckx VA, Naicker S. Acute kidney injury associated with the use of traditional medicines. *Nat Rev Nephrol*. 2008;4:664.
- Stanifer JW, Kilonzo K, Wang D, et al. Traditional medicines and kidney disease in low-and middle-income countries: opportunities and challenges. *Sem Nephrol*. 2017;37:245–259.
- Luyckx VA. Nephrotoxicity of alternative medicine practice. *Adv Chronic Kidney Dis*. 2012;19:129–141.
- Tannor EK, Hutton-Mensah KA. Naphthalene induced acute kidney injury in an African patient in Ghana: about a case report. *Pan Afr Med J*. 2019;32:201.
- Barr R, Rees P, Cordy P, Kungu A, Woodger B, Cameron H. Nephrotic syndrome in adult Africans in Nairobi. *Br Med J*. 1972;2:131–134.
- Olumide YM, Akinkugbe AO, Altraide D, et al. Complications of chronic use of skin lightening cosmetics. *Int J Dermatol*. 2008;47:344–353.
- Kibukamusoke J, Davies D, Hutt M. Membranous nephropathy due to skin-lightening cream. *Br Med J*. 1974;2:646.
- Suliman SM, Homeida M, Aboud OI. Paraphenylenediamine induced acute tubular necrosis following hair dye ingestion. *Hum Toxicol*. 1983;2:633–635.
- Warrell DA. Snake bite. *Lancet*. 2010;375:77–88.
- Chippaux JP. Snake-bites: appraisal of the global situation. *Bull World Health Organ*. 1998;76:515–524.
- Chippaux J-P, Goyffon M. Epidemiology of scorpionism: a global appraisal. *Acta Trop*. 2008;107:71–79.
- Isbister GK, Bawaskar HS. Scorpion envenomation. *New Engl J Med*. 2014;371:457–463.
- Olowu WA. Acute kidney injury in children in Nigeria. *Clin Nephrol*. 2015;83:70–74.
- Mehta RL, Cerda J, Burdmann EA, et al. International Society of Nephrology's 0by25 initiative for acute kidney injury (zero preventable deaths by 2025): a human rights case for nephrology. *Lancet*. 2015;385:2616–2643.
- Smoyer WE, Finkelstein FO, McCulloch MI, Carter M, Brusselmanns A, Feehally J. “Saving Young Lives” with acute kidney injury: the challenge of acute dialysis in low-resource settings. *Kidney Int*. 2016;89:254–256.
- Abdou N, Antwi S, Koffi LA, et al. Peritoneal dialysis to treat patients with acute kidney injury-the saving young lives experience in West Africa: proceedings of the saving young lives session at the first international conference of dialysis in West Africa, Dakar, Senegal, december 2015. *Perit Dial Int*. 2017;37:155–158.

51. McCulloch MI, Nourse P, Argent AC. Use of locally prepared peritoneal dialysis (PD) fluid for acute PD in children and infants in Africa. *Perit Dial Int*. 2020;40:441–445.
52. Ponce D, Balbi A, Cullis B. Acute PD: evidence, guidelines, and controversies. *Semin Nephrol*. 2017;37:103–112.
53. Cullis B, Abdelraheem M, Abrahams G, et al. Peritoneal dialysis for acute kidney injury. *Perit Dial Int*. 2014;34:494–517.
54. Naicker S, Aboud O, Gharbi MB. Epidemiology of acute kidney injury in Africa. *Semin Nephrol*. 2008;28:348–353.
55. Seedat YK, Nathoo BC. Acute renal failure in blacks and Indians in South Africa-comparison after 10 years. *Nephron*. 1993;64:198–201.
56. Dlamini TAL, Heering PJ, Chivese T, Rayner B. A prospective study of the demographics, management and outcome of patients with acute kidney injury in Cape Town, South Africa. *PLoS One*. 2017;12:e0177460.
57. Chothia MY, Moosa MR. Acute kidney injury outcomes at 90 days at a South African academic hospital. *Afr J Nephrol*. 2022;25:211–217.
58. Yao KH, Konan SD, Tia WM, Diopoh SP, Moh R, Sanogo S. Outcomes of acute kidney injury in a department of internal medicine in Abidjan (Cote d'Ivoire). *Nephrology (Carlton)*. 2018;23:653–660.
59. Hill NR, Fatoba ST, Oke JL, et al. Global prevalence of chronic kidney disease—a systematic review and meta-analysis. *PLoS One*. 2016;11:e0158765.
60. Kalyesubula R, Conroy AL, Calice-Silva V, et al. Screening for kidney disease in low-and middle-income countries. *Semin Nephrol*. 2022;151315. Elsevier.
61. Garcia-Garcia G, Jha V, Tao Li PK, et al. Chronic kidney disease (CKD) in disadvantaged populations. *Clin Kidney J*. 2015;8:3–6.
62. Cass A, Cunningham J, Snelling P, Wang Z, Hoy W. Exploring the pathways leading from disadvantage to end-stage renal disease for indigenous Australians. *Soc Sci Med*. 2004;58:767–785.
63. Tannor EK, Chika OU, Okpechi IG. The impact of low socioeconomic status on progression of chronic kidney disease in low-and lower middle-income countries. *Semin Nephrol*. 2022;42:151338.
64. Okpechi IG, Bello AK, Ameh OI, Swanepoel CR. Integration of care in management of CKD in resource-limited settings. *Semin Nephrol*. 2017;37:260–272.
65. Holness JL, van der Westhuizen DJ, Davids MR, Warwick JM. Measurement of glomerular filtration rate and its current status in African countries. *Afr J Nephrol*. 2022;25:74–81.
66. Fabian J. Estimating glomerular filtration rate in African populations. *Afr J Nephrol*. 2023;26:117–126.
67. van Deventer HE, George JA, Paiker JE, Becker PJ, Katz IJ. Estimating glomerular filtration rate in black South Africans by use of the Modification of Diet in Renal Disease and Cockcroft-Gault equations. *Clin Chem*. 2008;54:1197–1202.
68. Madala ND, Nkwanyana N, Dubula T, Naiker IP. Predictive performance of eGFR equations in South Africans of African and Indian ancestry compared with ^{99m}Tc-DTPA imaging. *Int Urol Nephrol*. 2012;44:847–855.
69. Moodley N, Hariparshad S, Peer F, Gounden V. Evaluation of the CKD-EPI creatinine based glomerular filtration rate estimating equation in Black African and Indian adults in KwaZulu-Natal, South Africa. *Clin Biochem*. 2018;59:43–49.
70. Seape T, Gounden V, van Deventer HE, Candy GP, George JA. Cystatin C-and creatinine-based equations in the assessment of renal function in HIV-positive patients prior to commencing highly active antiretroviral therapy. *Ann Clin Biochem*. 2016;53:58–66.
71. Holness JL, Bezuidenhout K, Davids MR, Warwick JM. Estimated glomerular filtration rate in a South African population of mixed ancestry. *S Afr Med J*. 2020;110:229–234.
72. Sagou Yayo E, Aye M, Konan J-L, et al. Inadequacy of the African-American ethnic factor to estimate glomerular filtration rate in an African general population: results from Côte d'Ivoire. *Nephrol Ther*. 2016;12:454–459.
73. Wyatt CM, Schwartz GJ, Owino Ong'or W, et al. Estimating kidney function in HIV-infected adults in Kenya: comparison to a direct measure of glomerular filtration rate by iothexol clearance. *PLoS One*. 2013;8:e69601.
74. Fabian J, Kalyesubula R, Mkandawire J, et al. Measurement of kidney function in Malawi, South Africa, and Uganda: a multicentre cohort study. *Lancet Global Health*. 2022;10:e1159–e1169.
75. Umeukeje EM, Koonce TY, Kusnoor SV, et al. Systematic review of international studies evaluating MDRD and CKD-EPI estimated glomerular filtration rate (eGFR) equations in Black adults. *PLoS One*. 2022;17:e0276252.
76. Fabian J, George JA, Etheredge HR, et al. Methods and reporting of kidney function: a systematic review of studies from sub-Saharan Africa. *Clin Kidney J*. 2019;12:778–787.
77. Benghanem Gharbi M, Elseviers M, Zamd M, et al. Chronic kidney disease, hypertension, diabetes, and obesity in the adult population of Morocco: how to avoid “over”- and “under”-diagnosis of CKD. *Kidney Int*. 2016;89:1363–1371.
78. Delanaye P, Glasscock RJ, De Broe ME. Epidemiology of chronic kidney disease: think (at least) twice. *Clin Kidney J*. 2017.
79. De Broe ME, Benghanem Gharbi M, Zamd M, Elseviers M. Why overestimate or underestimate chronic kidney disease when correct estimation is possible? *Nephrol Dial Transpl*. 2017;32:ii136–ii141.
80. Ekiti ME, Zambo J-B, Assah FK, Agbor VN, Kekay K, Ashuntantang G. Chronic kidney disease in sugarcane workers in Cameroon: a cross-sectional study. *BMC Nephrol*. 2018;19:10.
81. Stanifer JW, Jing B, Tolan S, et al. The epidemiology of chronic kidney disease in sub-Saharan Africa: a systematic review and meta-analysis. *Lancet Global Health*. 2014;2:e174–e181.
82. Diana NE, Naicker S. Update on current management of chronic kidney disease in patients with HIV infection. *Int J Nephrol Renovasc Dis*. 2016;9:223–234.
83. George C, Mogueo A, Okpechi I, Echouffo-Tcheugui JB, Kengne AP. Chronic kidney disease in low-income to middle-income countries: the case for increased screening. *BMJ Glob Health*. 2017;2:e000256.
84. Bonita R, de Courten M, Dwyer T, Jamrozik K, Winkelmann R. *Surveillance of Risk Factors for Noncommunicable Diseases: The WHO STEPwise Approach*. Geneva, Switzerland: World Health Organization; 2001.
85. Hariparshad S, Bhimma R, Nandlal L, Jembere E, Naicker S, Assounga A. The prevalence of chronic kidney disease in South Africa-limitations of studies comparing prevalence with sub-Saharan Africa, Africa, and globally. *BMC Nephrol*. 2023;24:62.
86. Kaze AD, Ilori T, Jaar BG, Echouffo-Tcheugui JB. Burden of chronic kidney disease on the African continent: a systematic review and meta-analysis. *BMC Nephrol*. 2018;19:125.
87. Abd EHS, Bolignano D, D'Arrigo G, Dounousi E, Tripepi G, Zoccali C. Prevalence and burden of chronic kidney disease among the general population and high-risk groups in Africa: a systematic review. *BMJ Open*. 2018;8:e015069.
88. Kayange NM, Smart LR, Tallman JE, et al. Kidney disease among children in sub-Saharan Africa: systematic review. *Pediatr Res*. 2014;77:272–281.
89. Freedman BI, Cohen AH. Hypertension-attributed nephropathy: what's in a name? *Nat Rev Nephrol*. 2016;12:27–36.
90. *IDF Diabetes Atlas 10th edition*. Brussels, Belgium: International Diabetes Federation; 2021.
91. Noubiap J, Naidoo J, Kengne AP. Diabetic nephropathy in Africa: a systematic review. *World J Diabetes*. 2015;6:759–773.
92. Wagnen F, Eshetie S, Kibret GD, et al. Diabetic nephropathy and hypertension in diabetes patients of sub-Saharan countries: a systematic review and meta-analysis. *BMC Res Notes*. 2018;11:1–7.
93. Ajayi SO, Ekrikpo UE, Ekanem AM, et al. Prevalence of chronic kidney disease as a marker of hypertension target organ damage in Africa: a systematic review and meta-analysis. *Int J Hypertens*. 2021;2021.
94. Davids MR, Marais N, Sebastian S, Jardine T, Jacobs JC. South African renal registry annual report 2021. *Afr J Nephrol*. 2023;26:83–94.
95. Boima V, Tannor EK, Osafo C, et al. The Ghana Renal Registry—a first annual report. *Afr J Nephrol*. 2021;24:19–24.
96. Hassaballa M, El-Wakil H, Elsharkawy M, et al. Egyptian renal data system (ERDS) 2020: an annual report of end-stage kidney disease patients on regular hemodialysis. *ESNT*. 2022;22:1–28.
97. Maritim PK, Twahir A, Davids MR. Global dialysis perspective: Kenya. *Kidney360*. 2022;3:1944–1947.
98. Niang A, Lemrabott AT. Global dialysis perspective: Senegal. *Kidney360*. 2020;1:538.
99. Milne F, James S, Veriava Y. Malignant hypertension and its renal complications in black South Africans. *SAMJ (S Afr Med J)*. 1989;78:164–167.

100. Meyers A, Whalley N, Martins M, Milne F, Margolius L. Hypertensive mechanisms in the indigenous (black) population of South Africa. *Nephrol*. 1998;4:S40–S47.
101. Katz IJ, Sofianou L, Butler O, Hopley M. Recovery of renal function in Black South African patients with malignant hypertension: superiority of continuous ambulatory peritoneal dialysis over hemodialysis. *Perit Dial Int*. 2001;21:581–586.
102. Gold CH, Isaacson C, Levin J. The pathological basis of end-stage renal disease in blacks. *S Afr Med J*. 1982;61:263–265.
103. Esmail AM, Bates WD, Amirali MH, Jardine T, Davids MR. Patterns of biopsy-proven kidney disease amongst South African adults from 1995 to 2017. *Afr J Nephrol*. 2023;26:9–16.
104. Vermeulen A, Menezes C, Mashabane M, et al. Patterns of renal disease: a 30-year renal biopsy study at Chris Hani Baragwanath academic hospital, Soweto, Johannesburg, South Africa. *S Afr Med J*. 2019;109:486–492.
105. Freedman BI, Murea M. Target organ damage in African American hypertension: role of APOL1. *Curr Hypertens Rep*. 2012;14:21–28.
106. Freedman BI, Langefeld CD. The new era of APOL1-associated glomerulosclerosis. *Nephrol Dial Transplant*. 2012;27:1288–1291.
107. Fogo A, Breyer JA, Smith MC, et al. Accuracy of the diagnosis of hypertensive nephrosclerosis in African Americans: a report from the African American Study of Kidney Disease (AASK) Trial. AASK pilot study investigators. *Kidney Int*. 1997;51:244–252.
108. Freedman BI, Sedor JR. Hypertension-associated kidney disease: perhaps no more. *J Am Soc Nephrol*. 2008;19:2047–2051.
109. Perneger TV, Whelton PK, Klag MJ, Rossiter KA. Diagnosis of hypertensive end-stage renal disease: effect of patient's race. *Am J Epidemiol*. 1995;141:10–15.
110. Schlessinger SD, Tankersley MR, Curtis JJ. Clinical documentation of end-stage renal disease due to hypertension. *Am J Kidney Dis*. 1994;23:655–660.
111. Okpechi IG, Ameh OI, Bello AK, Ronco P, Swanepoel CR, Kengne AP. Epidemiology of histologically proven glomerulonephritis in Africa: a systematic review and meta-analysis. *PLoS One*. 2016;11:e0152203.
112. Hargrey I, Coetzee L, Bates W, Chothia M-Y. C3 glomerulonephritis in Cape Town, South Africa—a case series. *Afr J Nephrol*. 2023;26:106–110.
113. Uwaezuoke SN, Ndu IK, Mbanefo NR. Prevalence rates of histopathologic subtypes associated with steroid resistance in childhood nephrotic syndrome in Sub-Saharan Africa: a systematic review. *Int J Nephrol Renovasc Dis*. 2019;167–176.
114. *In Danger: UNAIDS Global AIDS Update 2022*. Geneva, Switzerland: Joint United Nations Programme on HIV/AIDS; 2022.
115. *Global HIV & AIDS Statistics — Fact Sheet*. Geneva, Switzerland: UNAIDS; 2023.
116. *The Path that Ends AIDS: UNAIDS Global AIDS Update 2023*. Geneva, Switzerland: Joint United Nations Programme on HIV/AIDS; 2023.
117. Cohen SD, Kopp JB, Kimmel PL. Kidney diseases associated with human immunodeficiency virus infection. *N Engl J Med*. 2017;377:2363–2374.
118. Rao TK, Filippone EJ, Nicastrì AD, et al. Associated focal and segmental glomerulosclerosis in the acquired immunodeficiency syndrome. *N Engl J Med*. 1984;310:669–673.
119. Pardo V, Aldana M, Colton RM, et al. Glomerular lesions in the acquired immunodeficiency syndrome. *Ann Intern Med*. 1984;101:429–434.
120. Gardenswartz MH, Lerner CW, Seligson GR, et al. Renal disease in patients with AIDS: a clinicopathologic study. *Clin Nephrol*. 1984;21:197–204.
121. Hory B, Bresson C, Lorge J, Perol C. Associated focal and segmental glomerulosclerosis in the acquired immunodeficiency syndrome. *Am J Kidney Dis*. 1988;12:169.
122. Binda ki Muaka P, Pakasa-Muyulu N, Mbensa-Mansabi L. Abstracts of the 9th congress of the international pediatric Nephrology association. *Pediatr Nephrol*. 1992;6:C154.
123. Bates WD, Muller N, van de Wal BW, Jacobs JC. HIV-associated nephropathy—an initial presentation in an HIV-positive patient. *S Afr Med J*. 1994;84:223–224.
124. Fabian J, Naicker S. HIV and kidney disease in sub-Saharan Africa. *Nat Rev Nephrol*. 2009;5:591–598.
125. Rosenberg AZ, Naicker S, Winkler CA, Kopp JB. HIV-associated nephropathies: epidemiology, pathology, mechanisms and treatment. *Nat Rev Nephrol*. 2015;11:150–160.
126. Kopp JB, Nelson GW, Sampath K, et al. APOL1 genetic variants in focal segmental glomerulosclerosis and HIV-associated nephropathy. *J Am Soc Nephrol*. 2011;22:2129–2137.
127. Kasembeli AN, Duarte R, Ramsay M, et al. APOL1 risk variants are strongly associated with HIV-associated nephropathy in black South Africans. *J Am Soc Nephrol*. 2015.
128. Naicker S, Rahmanian S, Kopp JB. HIV and chronic kidney disease. *Clin Nephrol*. 2015;83:32–38.
129. Limou S, Nelson GW, Lecordier L, et al. Sequencing rare and common APOL1 coding variants to determine kidney disease risk. *Kidney Int*. 2015.
130. Koech M, Owiti M, Owino-Ong'or W, et al. Absence of HIV-associated nephropathy among antiretroviral naïve adults with persistent albuminuria in Western Kenya. *Kidney Int Rep*. 2017;2:159–164.
131. Han TM, Naicker S, Ramdial PK, Assounga AG. A cross-sectional study of HIV-seropositive patients with varying degrees of proteinuria in South Africa. *Kidney Int*. 2006;69:2243–2250.
132. Gerntholtz TE, Goetsch SJ, Katz I. HIV-related nephropathy: a South African perspective. *Kidney Int*. 2006;69:1885–1891.
133. Wearne N, Swanepoel CR, Boule A, Duffield MS, Rayner BL. The spectrum of renal histologies seen in HIV with outcomes, prognostic indicators and clinical correlations. *Nephrol Dial Transplant*. 2012;27:4109–4118.
134. Wearne N, Manning K, Price B, et al. The evolving spectrum of kidney histology in HIV-positive patients in South Africa. *Kidney Int Rep*. 2023;8:1087–1096.
135. Diana NE, Davies M, Mosiane P, Vermeulen A, Naicker S. Clinicopathological correlation of kidney disease in HIV infection pre- and post-ART rollout. *PLoS One*. 2022;17:e0269260.
136. Lucas GM, Eustace JA, Sozio S, Mentari EK, Appiah KA, Moore RD. Highly active antiretroviral therapy and the incidence of HIV-1-associated nephropathy: a 12-year cohort study. *AIDS*. 2004;18:541–546.
137. Szczech LA, Gupta SK, Habash R, et al. The clinical epidemiology and course of the spectrum of renal diseases associated with HIV infection. *Kidney Int*. 2004;66:1145–1152.
138. Reid A, Stohr W, Walker AS, et al. Severe renal dysfunction and risk factors associated with renal impairment in HIV-infected adults in Africa initiating antiretroviral therapy. *Clin Infect Dis*. 2008;46:1271–1281.
139. Peters PJ, Moore DM, Mermin J, et al. Antiretroviral therapy improves renal function among HIV-infected Ugandans. *Kidney Int*. 2008;74:925–929.
140. Mpondo BC, Kalluvya SE, Peck RN, et al. Impact of antiretroviral therapy on renal function among HIV-infected Tanzanian adults: a retrospective cohort study. *PLoS One*. 2014;9:e89573.
141. Fabian J, Naicker S, Goetsch S, Venter WD. The clinical and histological response of HIV-associated kidney disease to antiretroviral therapy in South Africans. *Nephrol Dial Transplant*. 2013;28:1543–1554.
142. Inusah A-J, Coetzee L, Bates W, Chothia M-Y. Kidney outcomes of immune-complex associated mesangiocapillary glomerulonephritis in patients with and without HIV. *J Nephropathol*. 2022;11.
143. López M, Hoppe B. History, epidemiology and regional diversities of urolithiasis. *Pediatr Nephrol*. 2010;25:49–59.
144. Watson G, Payne SR, Kunitzky K, Natchagande G, Mabedi C, Scotland KB. Stone disease in low- and middle-income countries: could augmented reality have a role in its management? *BJU Int*. 2022;130:400–407.
145. Sorokin I, Mamoulakis C, Miyazawa K, Rodgers A, Talati J, Lotan Y. Epidemiology of stone disease across the world. *World J Urol*. 2017;35:1301–1320.
146. Eisner BH, Sheth S, Herrick B, et al. The effects of ambient temperature, humidity and season of year on urine composition in patients with nephrolithiasis. *BJU Int*. 2012;110:E1014–E1017.
147. Collins AJ, Pitt B, Reaven N, et al. Association of serum potassium with all-cause mortality in patients with and without heart failure, chronic kidney disease, and/or diabetes. *Am J Nephrol*. 2017;46:213–221.

148. Kaestner L, Lazarus J, Muller E. Twenty-four-hour urine collection is appropriate in a cohort of South African renal stone formers. *Afr J Nephrol*. 2021;24:68–74.
149. Meiouet F, El Kabbaj S, Daudon M. Pediatric urolithiasis in Morocco: composition of 432 urinary calculi analyzed by infrared spectroscopy. *Prog Urol*. 2019;29:173–182.
150. Annee 2010. *Rapport D'Activite Des Centres et Unites D'Hemodialyse*. Tunisia: Ministère de la Santé Publique; 2011.
151. Banaga AS, Mohammed EB, Siddig RM, et al. Causes of end stage renal failure among haemodialysis patients in Khartoum State/Sudan. *BMC Res Notes*. 2015;8:1–7.
152. Elsharif ME, Elsharif EG. Causes of end-stage renal disease in Sudan: a single-center experience. *Saudi J Kidney Dis Transpl*. 2011;22:373–376.
153. Alashek WA, McIntyre CW, Taal MW. Epidemiology and aetiology of dialysis-treated end-stage kidney disease in Libya. *BMC Nephrol*. 2012;13:1.
154. Arogundade FA, Hassan MO, Omotoso BA, et al. Spectrum of kidney diseases in Africa: malaria, schistosomiasis, sickle cell disease, and toxins. *Clin Nephrol*. 2016;86:S53–S60.
155. Naik RP, Derebail VK, Grams ME, et al. Association of sickle cell trait with chronic kidney disease and albuminuria in African Americans. *JAMA*. 2014;312:2115–2125.
156. Nath KA, Heibel RP. Sickle cell disease: renal manifestations and mechanisms. *Nat Rev Nephrol*. 2015;11:161–171.
157. Ataga KI, Saraf SL, Derebail VK. The nephropathy of sickle cell trait and sickle cell disease. *Nat Rev Nephrol*. 2022;18:361–377.
158. Desai PC, Deal AM, Brittain JE, Jones S, Hinderliter A, Ataga KI. Decades after the cooperative study: a re-examination of systemic blood pressure in sickle cell disease. *Am J Hematol*. 2012;87:E65.
159. Benneh-Akwasi Kuma A, Owusu-Ansah AT, Ampomah MA, et al. Prevalence of relative systemic hypertension in adults with sickle cell disease in Ghana. *PLoS One*. 2018;13:e0190347.
160. Arogundade F, Sanusi A, Hassan M, Salawu L, Durosinmi M, Akinsola A. An appraisal of kidney dysfunction and its risk factors in patients with sickle cell disease. *Nephron Clin Pract*. 2011;118:c225–c231.
161. Orolu M, Mabayoje M, Braimoh R, et al. Iron indices in adults with sickle cell nephropathy in Lagos, Nigeria. *Afr J Nephrol*. 2022;25:67–173.
162. Kambale-Kombi P, Marini Djang'eing'a R, Opara J-PA, et al. Renal abnormalities among sickle cell disease patients in a poor management setting: a survey in the Democratic Republic of the Congo. *Mediterr J Hematol Infect Dis*. 2022;14.
163. Sasongko TH, Nagalla S. Angiotensin-converting enzyme (ACE) inhibitors for proteinuria and microalbuminuria in people with sickle cell disease. *Cochrane Database Syst Rev*. 2021.
164. Han J, Zhang X, Saraf SL, et al. Risk factors for vitamin D deficiency in sickle cell disease. *Br J Haematol*. 2018;181:828–835.
165. Jacob SA, Talati R, Kanter J. The evolving treatment landscape for children with sickle cell disease. *The Lancet Child & Adolescent Health*. 2023;7:797–808.
166. Nielsen L, Canoui-Poitrine F, Jais JP, et al. Morbidity and mortality of sickle cell disease patients starting intermittent haemodialysis: a comparative cohort study with non-Sickle dialysis patients. *Br J Haematol*. 2016;174:148–152.
167. Olaniran KO, Allegretti AS, Zhao SH, et al. Kidney function decline among black patients with sickle cell trait and sickle cell disease: an observational cohort study. *J Am Soc Nephrol*. 2020;31:393.
168. Kwarteng-Siaw M, Heydarpour M, Baker O, Tucker K, Achebe M. Morbidity and mortality associated with hemodialysis versus peritoneal dialysis in patients with end stage renal disease caused by sickle cell disease. *Blood*. 2021;138:488.
169. Huang E, Parke C, Mehrnia A, et al. Improved survival among sickle cell kidney transplant recipients in the recent era. *Nephrol Dial Transplant*. 2013;28:1039–1046.
170. H3Africa Consortium. Enabling the genomic revolution in Africa: H3Africa is developing capacity for health-related genomics research in Africa. *Science*. 2014;344:1346–1348.
171. Adedokun BO, Olopade CO, Olopade OI. Building local capacity for genomics research in Africa: recommendations from analysis of publications in Sub-Saharan Africa from 2004 to 2013. *Glob Health Action*. 2016;9:31026.
172. Ilori T, Watanabe A, Ng K-H, Solarin A, Sinha A, Gbadegesin R. Genetics of chronic kidney disease in low-resource settings. *Semin Nephrol*. 2022;42:151314.
173. Osafo C, Raji YR, Burke D, et al. Human Heredity and Health (H3) in Africa kidney disease research network: a focus on methods in sub-Saharan Africa. *Clin J Am Soc Nephrol*. 2015;10:2279–2287.
174. Osafo C, Raji YR, Olanrewaju T, et al. Genomic approaches to the burden of kidney disease in sub-Saharan Africa: the Human Heredity and Health in Africa (H3Africa) kidney disease research network. *Kidney Int*. 2016;90:2–5.
175. Adu D, Ojo A, H3A-KDRN Investigators. Overview of the Human Heredity and Health in Africa Kidney Disease Research Network (H3A-KDRN). *Kidney360*. 2021;2:129.
176. Jaouad IC, Elalaoui SC, Sbiti A, Elkerh F, Belmahi L, Sefiani A. Consanguineous marriages in Morocco and the consequence for the incidence of autosomal recessive disorders. *J Biosoc Sci*. 2009;41:575–581.
177. Bourquia A. Autosomal dominant polycystic kidney disease (AD-PKD) in Morocco. Multicenter study about 308 families [Fr]. *Nephrologie*. 2002;23:93–96.
178. Laleye A, Awede B, Agboton B, et al. Autosomal dominant polycystic kidney disease in University Clinic of Nephrology and Haemodialysis of Cotonou: clinical and genetical findings. *Genet Couns*. 2012;23:435–445.
179. Fary Ka E, Seck SM, Niang A, Cisse MM, Diouf B. Patterns of autosomal dominant polycystic kidney diseases in black Africans. *Saudi J Kidney Dis Transpl*. 2010;21:81–86.
180. Seck SM, Guèye S, Diouf B. A new PKD-1 mutation discovered in a black African woman with autosomal polycystic kidney disease. *Nephro-Urol Mon*. 2013;5:769–772.
181. Okyere P, Ephraim RK, Okyere I, et al. Demographic, diagnostic and therapeutic characteristics of autosomal dominant polycystic kidney disease in Ghana. *BMC Nephrol*. 2021;22:156.
182. Arogundade FA, Akinbodewa AA, Sanusi AA, Okunola O, Hassan MO, Akinsola A. Clinical presentation and outcome of autosomal dominant polycystic kidney disease in Nigeria. *Afr Health Sci*. 2018;18:671–680.
183. Yersin C, Bovet P, Wauters J, Schorderet D, Pescia G, Paccaud F. Frequency and impact of autosomal dominant polycystic kidney disease in the Seychelles (Indian Ocean). *Nephrol Dial Transplant*. 1997;12:2069–2074.
184. Lombard EH, Kromberg JG, Thomson PD, Milner LS, van Biljon I, Jenkins T. Autosomal recessive polycystic kidney disease. Evidence for high frequency of the gene in the Afrikaans-speaking population. *S Afr Med J*. 1989;76:321–323.
185. Lambie L, Amin R, Essop F, Cnaan A, Krause A, Guay-Woodford LM. Clinical and genetic characterization of a founder PKHD1 mutation in Afrikaners with ARPKD. *Pediatr Nephrol*. 2015;30:273–279.
186. Hildebrandt F, Zhou W. Nephronophthisis-associated ciliopathies. *J Am Soc Nephrol*. 2007;18:1855–1871.
187. Soliman NA, Hildebrandt F, Otto EA, et al. Clinical characterization and NPHP1 mutations in nephronophthisis and associated ciliopathies: a single center experience. *Saudi J Kidney Dis Transpl*. 2012;23:1090–1098.
188. Nagara M, Tiar A, Halim NB, et al. Mutation spectrum of primary hyperoxaluria type 1 in Tunisia: implication for diagnosis in North Africa. *Gene*. 2013;527:316–320.
189. Soliman NA, Nabhan MM, Abdelrahman SM, et al. Clinical spectrum of primary hyperoxaluria type 1: experience of a tertiary center. *Nephrol Ther*. 2017;13:176–182.
190. Owen EP, Nandhlal J, Leisegang F, Van der Watt G, Nourse P, Gajjar P. Common mutation causes cystinosis in the majority of black South African patients. *Pediatr Nephrol*. 2015;30:595–601.
191. El-Karakasy H, Fahmy M, El-Raziky M, et al. Hereditary tyrosinemia type 1 from a single center in Egypt: clinical study of 22 cases. *World J Pediatr*. 2011;7:224–231.
192. Chaabouni HB, Ksantini M, M'Rad R, et al. MEFV mutations in Tunisian patients suffering from familial Mediterranean fever. *Semin Arthritis Rheum*. 2007;36:397–401.
193. Ben-Chetrit E. Familial Mediterranean fever (FMF) and renal AA amyloidosis-phenotype-genotype correlation, treatment and prognosis. *J Nephrol*. 2003;16:431–434.
194. Al-Haggag M, Bakr A, Tajima T, et al. Familial hypomagnesemia with hypercalciuria and nephrocalcinosis: unusual clinical associations and novel claudin16 mutation in an Egyptian family. *Clin Exp Nephrol*. 2009;13:288–294.

195. van der Merwe PD, Rensburg MA, Haylett WL, Bardien S, Davids MR. Gitelman syndrome in a South African family presenting with hypokalaemia and unusual food cravings. *BMC Nephrol.* 2017;18:38.
196. Nkeh B, Samani NJ, Badenhorst D, et al. T594M variant of the epithelial sodium channel β -subunit gene and hypertension in individuals of African ancestry in South Africa. *Am J Hypertens.* 2003;16:847–852.
197. Jones ES, Owen EP, Rayner BL. The association of the R563Q genotype of the ENaC with phenotypic variation in Southern Africa. *Am J Hypertens.* 2012;25:1286–1291.
198. Dhanjal M, Owen E, Anthony J, Davidson J, Rayner B. Association of pre-eclampsia with the R563Q mutation of the β -subunit of the epithelial sodium channel. *BJOG.* 2006;113:595–598.
199. Elhayek D, de Nanclares GP, Chouchane S, et al. Molecular diagnosis of distal renal tubular acidosis in Tunisian patients: proposed algorithm for Northern Africa populations for the ATP6V1B1, ATP6V0A4 and SCL4A1 genes. *BMC Med Genet.* 2013;14:119.
200. Lenkkeri U, Männikkö M, McCready P, et al. Structure of the gene for congenital nephrotic syndrome of the Finnish type (NPHS1) and characterization of mutations. *Am J Hum Genet.* 1999;64:51–61.
201. Gee HY, Saisawat P, Ashraf S, et al. ARHGDI mutations cause nephrotic syndrome via defective RHO GTPase signaling. *J Clin Invest.* 2013;123:3243–3253.
202. Nandlal L, Winkler CA, Bhimma R, et al. Causal and putative pathogenic mutations identified in 39% of children with primary steroid-resistant nephrotic syndrome in South Africa. *Eur J Pediatr.* 2022;181:3595–3606.
203. Abid S, Hassen W, Achour A, et al. Ochratoxin A and human chronic nephropathy in Tunisia: is the situation endemic? *Hum Exp Toxicol.* 2003;22:77–84.
204. Stanifer JW, Maro V, Egger J, et al. The epidemiology of chronic kidney disease in Northern Tanzania: a population-based survey. *PLoS One.* 2015;10:e0124506.
205. Kamel EG, El-Minshawy O. Environmental factors incriminated in the development of end stage renal disease in El-Minia Governorate, Upper Egypt. *Int J Nephrol Urol.* 2010;2:431–437.
206. Hamilton SA, Nakanga WP, Prynn JE, et al. Prevalence and risk factors for chronic kidney disease of unknown cause in Malawi: a cross-sectional analysis in a rural and urban population. *BMC Nephrol.* 2020;21:1–12.
207. Ashuntantang G, Osafo C, Olowu WA, et al. Outcomes in adults and children with end-stage kidney disease requiring dialysis in sub-Saharan Africa: a systematic review. *Lancet Glob Health.* 2017;5:e408–e417.
208. Ashuntantang G, Miljeteig I, Luyckx VA. Bedside rationing and moral distress in nephrologists in sub-Saharan Africa. *BMC Nephrol.* 2022;23:1–14.
209. Wearne N, Krause R, Davidson B, Brennan F. Renal palliative and supportive care in South Africa—a consensus statement. *Afr J Nephrol.* 2020;23:86–107.
210. Davids MR, Eastwood JB, Selwood NH, et al. A renal registry for Africa: first steps. *Clin Kidney J.* 2016;9:162–167.
211. Hole BD, Evans KM, Pyart R, et al. International collaborative efforts to establish kidney health surveillance systems. *Kidney Int.* 2020;98:812–816.
212. Jardine T, Wong E, Steenkamp R, Caskey FJ, Davids MR. Survival of South African patients on renal replacement therapy. *Clin Kidney J.* 2020;13:782–790.
213. Thapa S, Jardine T, Davids T, Caskey FJ, Davids MR. Incidence and one-year survival of elderly South Africans starting kidney replacement therapy. *Kidney Int Rep.* 2022;7:2071–2075.
214. Muller E, Barday Z, Mendelson M, Kahn D. HIV-positive-to-HIV-negative kidney transplantation—results at 3 to 5 years. *N Engl J Med.* 2015;372:613–620.
215. Banda J, Bulaya T, Mweemba A, et al. Early mortality in patients with kidney failure starting chronic dialysis in Zambia: a retrospective cohort analysis. *Afr J Nephrol.* 2022;25:3–7.
216. Teuwafu DG, Mambap AT, Alake PA, Gobina RM, Halle M-P, Ashuntantang G. Survival in elderly patients with kidney failure starting haemodialysis in Cameroon. *Afr J Nephrol.* 2023;26:111–116.
217. Okaka EI, Naidoo S, Ahmed MM, Davies M, Naicker S. Quality of life in patients on continuous ambulatory peritoneal dialysis in an African setting. *Saudi J Kidney Dis Transpl.* 2015;26:631–637.
218. Tannor EK, Archer E, Kapembwa K, Van Schalkwyk SC, Davids MR. Quality of life in patients on chronic dialysis in South Africa: a comparative mixed methods study. *BMC Nephrol.* 2017;18:4.
219. Masina T, Chimera B, Kamponda M, Dreyer G. Health related quality of life in patients with end stage kidney disease treated with haemodialysis in Malawi: a cross sectional study. *BMC Nephrol.* 2016;17:61.
220. Kamau E, Kayima J, Otieno C, Maritim M, Wanzala P. Health related quality of life of patients on maintenance haemodialysis at Kenyatta National Hospital. *East Afr Med J.* 2012;89:75–81.
221. Msilanga DP, Punatar P, Ruggajo P. Health-related quality of life of patients undergoing haemodialysis therapy in Dar es Salaam, Tanzania. *Afr J Nephrol.* 2023;26:70–75.
222. Bagasha P, Naitala R, Namukwaya E, Leng M, Katabira E, Namisango E. Cultural adaptation and validation of the Kidney Disease and quality of Life-Short Form (KDQOL-SF™) version 1.3 questionnaire in Uganda. *Afr J Nephrol.* 2022;25:116–135.
223. Bagasha P, Namukwaya E, Leng M, et al. Comparison of the health-related quality of life of end stage kidney disease patients on hemodialysis and non-hemodialysis management in Uganda. *BMC Palliat Care.* 2021;20:1–13.
224. El Filali A, Bentata Y, Ada N, Oneib B. Depression and anxiety disorders in chronic hemodialysis patients and their quality of life: a cross-sectional study about 106 cases in the northeast of Morocco. *Saudi J Kidney Dis Transpl.* 2017;28:341–348.
225. Boudida B, Rhou H, Ezaitouni F, et al. Translation, cultural adaptation and validation of the kidney disease quality of life—short form 1.3 in an African country. *Transpl Proc.* 2014;46:1295–1301.
226. Abdelfatah M, Shawky S, Azzab E. Quality of life in dialysis versus kidney transplantation. *Afr J Nephrol.* 2010;14:52–60.
227. Gbaguidi GN, Houehanou CY, Amidou SA, Vigan J, Houinato DS, Lacroix P. Prevalence of abnormal kidney function in a rural population of Benin and associated risk factors. *BMC Nephrol.* 2021;22:1–9.
228. George JA, Brandenburg J-T, Fabian J, et al. Kidney damage and associated risk factors in rural and urban sub-Saharan Africa (AWI-Gen): a cross-sectional population study. *Lancet Glob Health.* 2019;7:e1632–e1643.
229. Kaze FF, Maimouna M, Beybey AF, et al. Prevalence and determinants of chronic kidney disease in urban adults' populations of northern Cameroon. *Clin Exp Nephrol.* 2021;25:718–726.
230. Masimango M, Sumaili E, Wallemacq P, Malembaka E, Hermans M, Fillée C. Prevalence and risk factors of chronic kidney disease in South-Kivu, Democratic Republic of Congo. *Kidney Int Rep.* 2020;5:1251–1260.
231. Olanrewaju TO, Aderibigbe A, Popoola AA, et al. Prevalence of chronic kidney disease and risk factors in North-Central Nigeria: a population-based survey. *BMC Nephrol.* 2020;21:1–10.
232. Wokoma FS, Emem-Chioma PC, Oko-Jaja RI. Systematic analysis of community studies of risk factors and prevalence of chronic kidney disease in Nigeria (2006–2014). *Trop J Nephrol.* 2017;11:7–16.
233. Adeniyi AB, Laurence CE, Volmink JA, Davids MR. Prevalence of chronic kidney disease and association with cardiovascular risk factors among teachers in Cape Town, South Africa. *Clin Kidney J.* 2017;10:363–369.
234. Hodel NC, Hamad A, Praehauser C, et al. The epidemiology of chronic kidney disease and the association with non-communicable and communicable disorders in a population of sub-Saharan Africa. *PLoS One.* 2018;13:e0205326.
235. Plath DW, Mbawambo JK, Fonner VA, et al. Prevalence of CKD, diabetes, and hypertension in rural Tanzania. *Kidney Int Rep.* 2018;3:905–915.
236. Muir AN, Charlebois ED, Balzer LB, et al. The epidemiology of chronic kidney disease (CKD) in rural East Africa: a population-based study. *PLoS One.* 2020;15:e0229649.
237. Kalyesubula R, Nankabirwa JI, Ssinabulya I, et al. Kidney disease in Uganda: a community based study. *BMC Nephrol.* 2017;18:116.
238. Sumaili E, Ekulu P, Pakasa N, Tshala-Katumbay D, Nseka N. Nephrology in the democratic republic of the Congo. In: Moura-Neto JA, Divino-Filho JC, Ronco C, eds. *Nephrology Worldwide*. Springer Nature Switzerland; 2021:25–34.
239. Mengistu YT, Ejigu AM. Global dialysis perspective: Ethiopia. *Kidney360.* 2022;3:1431–1434.

240. Okyere I, Okyere P, Ephraim RKD, et al. Vascular access for haemodialysis in Ghana: a single centre experience. *PAMJ Clinical Medicine*. 2021;6:2.
241. Abdulla A, Shiham I, Davids MR. Global dialysis perspective: Maldives. *Kidney360*. 2023;10:34067.
242. Wan DIM, Fagoonee K. Global dialysis perspective: Mauritius. *Kidney360*. 2021;2:1021–1026.
243. Okoye O, Mamven M. Global dialysis perspective: Nigeria. *Kidney360*. 2022;3:1607–1610.
244. Faye M, Raqui I, Faye M, et al. Vascular access in Senegalese patients starting chronic haemodialysis. *Afr J Nephrol*. 2023;26:3–8.
245. Kalyesubula R, Brewster U, Kansime G. Global dialysis perspective: Uganda. *Kidney360*. 2022;3:933–936.